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FEATURE-FUSION TECHNIQUES FOR BREAST MICROCALCIFICATION CLASSIFICATION - ACROSS 2D MAMMOGRAPHY, 3D MICRO-CT AND TABULAR DATA

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Abstract

Deep learning (DL) has been increasingly applied to breast cancer detection and classification [1]. In this context, breast microcalcifications (MCs) are an important imaging finding associated with breast cancer and serve as a primary indicator for early detection [2]. Clinical screening relies heavily on in-vivo 2D mammography for initial assessment [3, 2]. For further evaluation and research, 3D imaging is often utilized, ranging from in-vivo digital breast tomosynthesis (DBT) [4] to high-resolution, ex-vivo Micro-Computed Tomography (Micro-CT) of biopsy specimens [5]. Regardless of the modality, clinical diagnosis benefits from combining these visual findings with structured tabular information, such as patient demographics or quantitative radiomic features [6]. Integrating these different data types is not straightforward and remains a challenge, as naive fusion methods frequently overemphasize the high-dimensional image data at the expense of tabular features, a phenomenon known as modality dominance [7].

In this thesis, we investigate two multimodal feature-fusion strategies for breast MC classification: decision-level late fusion and the Dynamic Affine Feature Map Transform (DAFT) [8], an intermediate feature-fusion approach that uses tabular features to dynamically condition intermediate image feature maps inside the neural network. These approaches are evaluated in (1) a public 2D mammography dataset and (2) an in-house high-resolution 3D Micro-CT dataset of individual breast MCs.

The first pipeline utilized 455 patients from the EMory BrEast imaging Dataset (EMBED) [12], combining 1,277 2D mammography regions of interest (ROIs) with clinical tabular data. In this dataset, the image-only and tabular-only baselines yielded limited performance. The intermediate DAFT approach improved upon these unimodal baselines, achieving a 5-fold cross-validation (CV) mean Area Under the Curve (AUC) of 0.6750.

The second pipeline evaluated individual 3D MCs paired with handcrafted radiomic features [18]. In this setting, the tabular-only baseline classifier (Extreme Gradient Boosting, or XGBoost) yielded an AUC of 0.8397, outperforming the image-only 3D convolutional neural network (CNN) baseline (AUC 0.6468). While 3D DAFT improved upon the image-only model (AUC 0.8062), an ablation study showed that restricting the tabular input to the top 200 radiomic features helped achieve an improved AUC of 0.8328. The decision-level late fusion approach yielded an overall AUC of 0.8439. Patient-clustered bootstrapping analysis indicated no statistically significant difference in AUC between the multimodal pipelines and the tabular-only baseline ($p = 0.9414$).

These results suggest that radiomic features provide a strong baseline for 3D MC classification. Combining high-dimensional 3D visual representations with radiomic descriptors did not yield statistically significant improvements in continuous risk-ranking or absolute binary classification accuracy, suggesting that the deep visual features act primarily as redundant information in this specific ex-vivo environment.

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Disclosure on the use of AI

Declaration

During the preparation of this work, the author used the following tool and service:

Name of tool/service	Reason
<i>Gemini (Google)</i>	in order to: assist with language editing, improve the clarity and structure of the written text, support code debugging and development, and discuss the interpretation and presentation of experimental results.

After using this tool, the author reviewed and edited the generated content and independently verified the code, analyses, results, and interpretations included in the thesis. The author takes full responsibility for the final content of this work.

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List of Abbreviations

AUC	Area Under the Curve
BI-RADS	Breast Imaging Reporting and Data System
CAD	Computer-Aided Diagnosis
CADe	Computer-Aided Detection
CI	Confidence Interval
CLAHE	Contrast-Limited Adaptive Histogram Equalization
CNN	Convolutional Neural Network
CV	Cross-Validation
DAFT	Dynamic Affine Feature Map Transform
DBT	Digital Breast Tomosynthesis
DCE-MRI	Dynamic Contrast-Enhanced Magnetic Resonance Imaging
DL	Deep Learning
EHR	Electronic Health Records
EMBED	EMory BrEast imaging Dataset
FC	Fully Connected (Layer)
FiLM	Feature-wise Linear Modulation
GAP	Global Average Pooling
GLCM	Gray Level Co-occurrence Matrix
GLRLM	Gray Level Run Length Matrix
MC	Microcalcification
Micro-CT	Micro-Computed Tomography
MRI	Magnetic Resonance Imaging
ReLU	Rectified Linear Unit
ROC	Receiver Operating Characteristic
ROI	Region of Interest
SVM	Support Vector Machine
XGBoost	Extreme Gradient Boosting

Chapter 1

Introduction

1.1 Context and Motivation

Breast cancer is one of the most frequently diagnosed cancers and a leading cause of cancer-related mortality among women worldwide. Early detection is paramount to improving survival rates, minimizing treatment invasiveness, and lowering healthcare expenditures. Screening mammography remains the standard clinical imaging modality for early detection [3]. Among various radiological biomarkers, breast microcalcifications (MCs)—small calcium deposits within mammary soft tissues—serve as critical early indicators of potential malignancy [2].

Despite technological advancements, conventional 2D mammography suffers from structural superposition and limited spatial resolution [3]. Overlapping fibroglandular tissue frequently obscures subtle MCs or creates artifactual patterns that mimic malignant lesions. Consequently, radiologists encounter diagnostic ambiguity, leading to false-positive findings that trigger unnecessary minimally invasive breast biopsies, or false-negative diagnoses where early-stage carcinomas pass undetected. To support radiological assessment, Computer-Aided Diagnosis (CAD) systems are developed to provide secondary computational analysis of medical images.

To overcome the physical imaging limitations for research and CAD development, high-resolution Micro-Computed Tomography (Micro-CT) is utilized. The clinical workflow begins with 2D mammography; if a suspicious area is marked, the patient undergoes a biopsy [18]. These extracted biopsy specimens are then scanned ex-vivo using Micro-CT at microscopic resolutions (e.g., $9\mu\text{m}$), providing an isotropic, tissue-superimposition-free 3D representation of the MCs [18]. However, clinical diagnosis relies on more than visual data alone; physicians inherently combine imaging findings with structured patient context, including age, family history, hormonal status, and quantitative radiomic descriptors.

Recent advancements in CAD rely heavily on deep learning (DL) to automatically extract hierarchical visual features from medical images. Concurrently, the field of radiomics - the extraction of explicitly defined mathematical descriptors from regions of interest - provides highly structured, quantitative tabular data. Because modern clinical diagnosis naturally synthesizes both image-derived visual data and this tabular/radiomic information, multimodal fusion techniques have emerged to integrate these heterogeneous data types into unified predictive models. These fusion strategies are generally categorized into three levels: early (input) fusion, which combines raw data formats prior to feature extraction; intermediate (feature-level) fusion, which integrates data streams within the internal layers of a neural network; and late (decision-level) fusion, which aggregates the final predictions of separate, independently trained unimodal networks.

1.2 Problem Statement

Combining high-dimensional visual tensor data (2D mammography or 3D Micro-CT images) with low-dimensional tabular features (clinical parameters and radiomics) introduces several algorithmic challenges [1]:

1. **Dimensionality Imbalance:** Images contain tens of thousands of continuous spatial pixels/voxels, whereas tabular data vectors typically consist of tens or hundreds of discrete variables. In naive joint networks, the optimizer often prioritizes high-dimensional visual gradients, underutilizing tabular features.
2. **Cross-Modal Feature Interaction:** Standard intermediate fusion techniques rely on simple concatenation prior to final dense layers. This late-stage interaction prevents clinical features from influencing early and intermediate spatial feature extraction inside the neural network backbone [8].
3. **Modality Disparity:** While DL generally establishes state-of-the-art performance in large-scale medical imaging, radiomics can provide highly competitive baselines in certain data-constrained settings, such as analyzing sparse individual MCs [10, 11]. When a significant performance gap exists between the modalities, standard decision-level ensembles can struggle to synthesize them without the less accurate modality reducing the overall predictive capability.
4. **Out-of-Distribution Vulnerability:** Purely tabular models are susceptible to overfitting small dataset splits or highly correlated feature spaces, leading to performance degradation when exposed to anomalous patient demographics or noisy high-dimensional radiomics.

1.3 Research Objectives

The objective of this thesis is to evaluate multimodal fusion architectures across two distinct experimental pipelines. The first pipeline evaluates a public dataset utilizing clinical 2D mammography combined with structured clinical and demographic tabular data. The second pipeline evaluates an in-house dataset of individual breast MCs, utilizing high-resolution 3D Micro-CT—an ex-vivo research technique—combined with radiomic features.

The primary research objectives are to:

- **Objective 1:** Implement unimodal baseline models (image-only Convolutional Neural Networks (CNNs), ResNet-18 architectures, and tabular-only models like Support Vector Machines (SVM) and Extreme Gradient Boosting (XGBoost)) to establish benchmark classification performance limits for both 2D mammography and 3D Micro-CT.
- **Objective 2:** Evaluate decision-level (late) fusion against feature-level (intermediate) fusion via the Dynamic Affine Feature Map Transform (DAFT) [8] to compare the efficacy of dynamically modifying intermediate image feature maps using tabular data versus aggregating final independent predictions.
- **Objective 3:** Evaluate the generalizability of multimodal architectures using patient-based cross-validation (CV), and assess the impact of feature space reduction through radiomic ablation studies.

Chapter 2

State of the Art

2.1 Breast Imaging Modalities: From Clinical Screening to Ex-Vivo Research

Screening programs are vital for the early detection of breast cancer, aiming to reduce morbidity, mortality, and treatment costs by identifying tumors before they become clinically symptomatic [13, 14]. The physical constraints and objectives of medical imaging differ drastically depending on whether the procedure is performed *in vivo* (on a living patient) or *ex vivo* (on extracted tissue samples for research).

2.1.1 In-Vivo Clinical Imaging: 2D Mammography

The current standard for global breast cancer screening is 2D mammography, which utilizes low-dose X-rays to capture structural images of the breast [3]. Because it is highly scalable, cost-effective, and standardized, it serves as the foundational imaging modality for both clinical workflows and large-scale public datasets (such as the EMBED dataset evaluated in this thesis).

However, traditional 2D mammography has a well-documented physical limitation: projecting a 3D volume onto a 2D plane causes the superimposition of fibroglandular tissues [15]. This overlapping can obscure subtle, early-stage MCs or create artifactual patterns that mimic cancer, contributing to diagnostic ambiguity [15, 2]. To partially mitigate this superimposition in clinical environments, Digital Breast Tomosynthesis (DBT) is sometimes employed to capture multiple X-ray projections and reconstruct semi-3D slices [4]. While DBT provides additional context, conventional 2D mammography remains the dominant screening tool and the primary focus of clinical CAD development in this work.

2.1.2 Ex-Vivo Research Imaging: Micro-CT

While clinical modalities must strictly limit radiation exposure, pre-clinical research can utilize imaging techniques that prioritize maximum spatial resolution. Micro-CT is a broad, non-destructive 3D imaging technology utilized across various fields of materials science and biomedical research [17].

Due to the exceptionally high radiation dose required to achieve microscopic resolution (e.g., $9\mu\text{m}$), Micro-CT is fundamentally unsuitable for *in vivo* clinical screening or patient diagnostics [17]. Instead, within the context of breast cancer research, it serves strictly as an *ex vivo* tool

applied to small, excised biopsy specimens [16]. By rotating the tissue sample, Micro-CT generates isotropic, tissue-superimposition-free volumetric representations. In this thesis, it provides a highly detailed 3D environment for extracting quantitative radiomic features from individual MCs, bypassing the visual limitations of in-vivo mammography.

2.2 Microcalcifications as Clinical Biomarkers

MCs are tiny calcium deposits within the soft tissues of the breast and represent one of the most critical early radiological indicators of breast cancer [2, 19]. Differentiating between benign and malignant MCs is a highly complex diagnostic task that relies on several primary characteristics:

- **Morphology:** The shape of an MC is a strong indicator of its origin. Benign calcifications tend to have smooth, uniform shapes, and are often described as round, coarse, eggshell-like, or teacup-shaped [19]. Conversely, malignant MCs are typically irregular. They are frequently classified as fine pleomorphic (resembling crushed stones of varying shapes) or fine-linear and branching [19].
- **Distribution Patterns:** The spatial arrangement of MCs within the breast offers important diagnostic clues. According to the Breast Imaging Reporting and Data System (BI-RADS), a globally standardized risk assessment and reporting framework developed by the American College of Radiology, distributions can be grouped, regional, diffuse, segmental, or linear [19]. Linear and segmental arrangements strongly suggest that the calcium deposits follow the ductal anatomy of the breast, carrying a high probability of malignancy [19].
- **Biochemical Composition:** MCs can also be differentiated chemically. Type I calcifications consist of calcium oxalate and are predominantly found in benign lesions [20]. Type II calcifications consist of hydroxyapatite and are frequently associated with malignant tumors, as the microenvironment of rapid cellular necrosis promotes this specific crystallization [20].

2.3 Deep Learning and Radiomics for Microcalcification Analysis

Historically, CAD systems for breast cancer focused on identifying *clusters* of MCs on 2D mammograms. Modern DL architectures, particularly CNNs like ResNet, have achieved state-of-the-art performance in this domain, frequently rivaling human radiologist performance in large-scale screening evaluations [22]. The development of these robust visual models has been heavily facilitated by massive public datasets, such as the EMBED dataset, the open-source part of which is utilized in this thesis [12]. Recent baseline validations utilizing deep CNNs on the EMBED dataset have demonstrated highly competitive performance, with researchers reporting AUC scores ranging from 0.70 to over 0.80 for automated abnormality classification and multi-year risk prediction [12].

However, while end-to-end CNNs excel at extracting hierarchical visual features from large repositories of 2D clinical images, analyzing the 3D micro-morphology of individual MCs introduces unique challenges. When operating in data-constrained scenarios (such as highly specialized pre-clinical cohorts), or when explicit mathematical interpretability is required to justify clinical decisions, researchers utilize radiomics as a powerful complementary approach to DL.

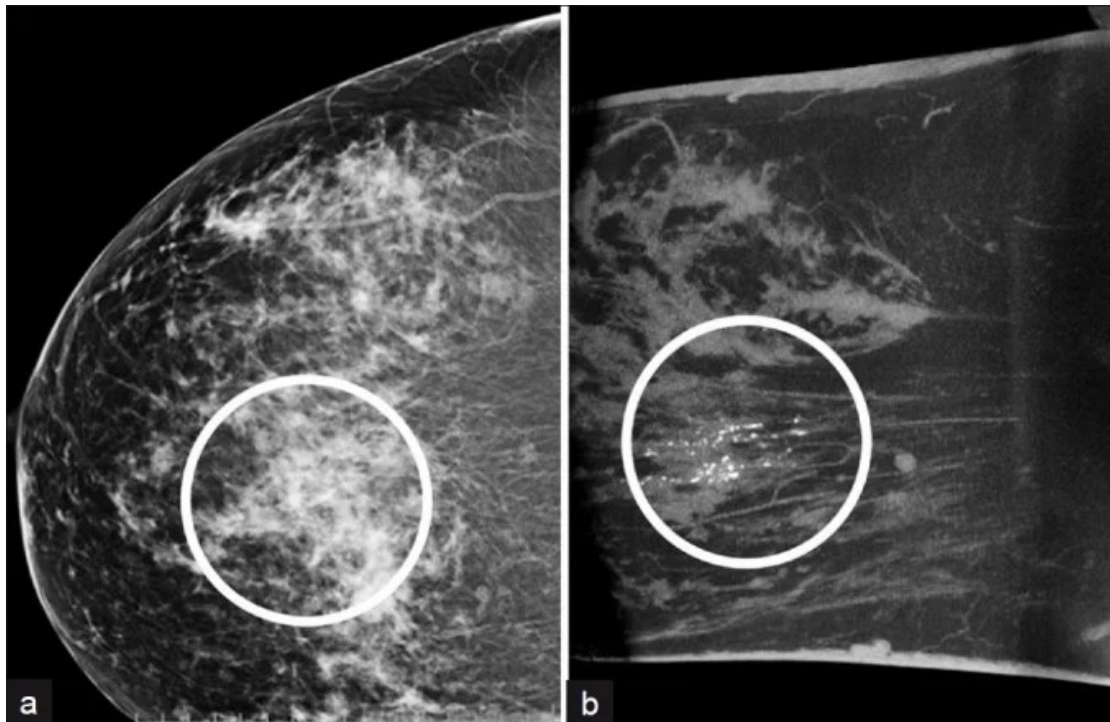


Figure 2.1: A fifty-five-year-old woman with MCs in a screening mammogram. (a) Diagnostic RCC demonstrates clusters of MCs within overlapping glandular tissue. Despite their small size, these calcifications are more conspicuous in (b) volume-rendered images at a standard 0.27-mm^3 resolution. Stereotactic biopsy demonstrated DCIS. Reproduced from O’Connell et al. [35].

Radiomics is the computational process of extracting massive amounts of explicitly defined, quantitative mathematical descriptors from medical images [23]. In the context of individual MCs, these features include:

- **First-Order Statistics:** Metrics describing the distribution of voxel intensities, such as mean, variance, and entropy.
- **Shape-Based Features:** Descriptors of physical dimensions, such as 3D volume, maximum 3D diameter, sphericity, and elongation.
- **Texture Features:** Complex, second-order statistics that describe the spatial relationships of pixel intensities. Common matrices include the Gray Level Co-occurrence Matrix (GLCM) and Gray Level Run Length Matrix (GLRLM), which effectively map the internal heterogeneity of malignant tissue [23].

Extensive research has applied these radiomic principles specifically to the high-resolution 3D Micro-CT imaging of breast biopsies. Previous work utilizing the same in-house Micro-CT dataset evaluated in this thesis demonstrated that explicitly engineered 3D radiomics can effectively capture the morphological complexity of individual MCs [18]. Building upon this framework, subsequent research introduced erosion-dilation trajectory radiomics to further analyze these 3D high-resolution MCs [24]. Furthermore, a 2026 analysis of this Micro-CT dataset

revealed that the tissue interface immediately surrounding the MC serves as a diagnostic epicenter, showing that radiomic features extracted from these boundary regions improve the prediction of malignancy [25].

Traditionally, these handcrafted radiomic features are passed into machine learning algorithms like SVM or XGBoost. However, to maximize diagnostic accuracy, an active area of CAD research involves multimodal fusion: attempting to dynamically integrate the automated spatial feature extraction of deep CNNs with the structured, clinically interpretable tabular data provided by radiomics and patient demographics [6, 8].

2.4 Taxonomy of Multimodal Fusion Techniques

Modern clinical diagnosis relies on the synthesis of high-dimensional visual data and low-dimensional tabular data (e.g., patient demographics, genetic markers, and radiomic features) [1]. DL models designed for this task must systematically fuse these heterogeneous structures [1]. These strategies are broadly classified by the depth at which the data streams intersect [1]:

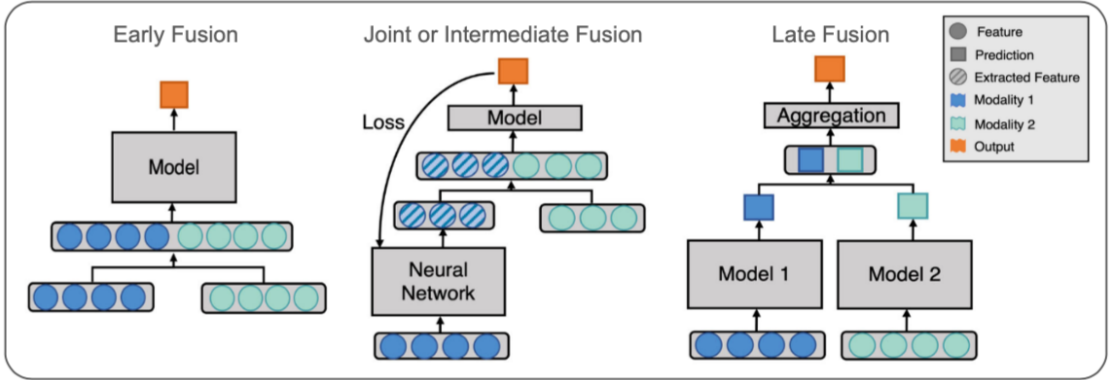


Figure 2.2: A taxonomy of multimodal fusion architectures. Early Fusion (left) combines raw modalities immediately at the input level prior to extraction. Joint or Intermediate Fusion (middle) merges representations extracted from deeper layers of the neural network, allowing modalities to interact during feature learning. Late Fusion (right) processes each modality independently and aggregates their final predictive outputs. Reproduced from SLDS LMU [36].

2.4.1 Input Fusion (Early Fusion)

Raw data from various modalities are aligned and concatenated before entering a single neural network backbone [1]. Because early fusion operates by concatenating raw data, it naturally requires inputs to share identical tensor dimensions (e.g., co-registered PET and CT images) [1]. Consequently, it is generally not applicable for fusing high-dimensional image volumes with 1D tabular data.

2.4.2 Output Fusion (Late Fusion)

Output fusion handles each modality independently, extracting predictions via specialized unimodal networks [1]. These predictions are then aggregated using decision-level mechanisms like weighted voting, averaging, or a secondary meta-classifier (e.g., Logistic Regression) [1]. While

this prevents the network from learning feature-level synergies across modalities, it allows researchers to leverage powerful, independently optimized architectures for both vision and tabular tasks [1].

2.4.3 Intermediate Fusion (Feature-Level Fusion)

This approach utilizes separate neural branches to extract intermediate features from each modality, subsequently merging them inside the network prior to final classification [1]. Simple intermediate fusion involves flattening a CNN’s feature map and concatenating it with a tabular vector [8]. However, this often causes a parameter imbalance, leading the network to overfit the massive visual feature space while ignoring the lower-dimensional clinical data [8].

2.5 Related Work in Multimodal Fusion for Breast Cancer

Recent research in breast cancer diagnosis has explored multimodal fusion to address the performance limitations of unimodal models. Current architectures increasingly combine imaging data (Mammography, Dynamic Contrast-Enhanced Magnetic Resonance Imaging (DCE-MRI), or Ultrasound) with clinical Electronic Health Records (EHR) or radiomic signatures.

2.5.1 Fusing Mammography with Clinical Data

In clinical screening workflows, radiologists interpret 2D mammograms alongside a patient’s clinical history. Researchers have modeled this process by combining mammography CNNs with tabular clinical risk factors (e.g., age, breast density, and family history). For example, Yala et al. developed a hybrid DL model integrating full-field digital mammograms with traditional clinical risk factors [26]. This multimodal approach improved upon the clinical-only Tyrer-Cuzick model for 5-year breast cancer risk prediction, raising the AUC from 0.62 to 0.70 [26]. These findings indicate that tabular data provides context complementary to image features.

2.5.2 Fusing Deep Learning with Radiomics in MRI and Ultrasound

Beyond mammography, multimodal fusion has been applied to Magnetic Resonance Imaging (MRI) and Ultrasound to capture varying tumor characteristics. Studies suggest that combining explicitly defined radiomic features with deep convolutional features can improve diagnostic performance. For instance, Antropova et al. evaluated a fusion methodology across breast MRI, Ultrasound, and Mammography datasets [27]. By integrating intermediate CNN-extracted features with radiomics and clinical data, their fusion model increased the classification AUC on the DCE-MRI dataset from 0.82 (CNN alone) to 0.89 (fused), further supporting the complementary nature of these feature spaces [27].

2.5.3 Intermediate versus Late Fusion Architectures

A key methodological consideration in multimodal integration is the fusion strategy. Often, researchers utilize decision-level (late) fusion, wherein an image-only CNN and a tabular-only model (e.g., XGBoost) are trained independently, and their final probabilities are averaged or ensembled. While structurally simple, late fusion prevents the tabular data from influencing the spatial feature extraction process [1].

Consequently, studies have increasingly investigated feature-level (intermediate) fusion. Intermediate architectures allow tabular data to interact with visual feature maps inside the CNN

backbone prior to the final prediction. One such method is the DAFT architecture introduced by Wolf et al. [8]. By utilizing tabular data to scale and shift the intermediate spatial feature maps of a CNN, DAFT demonstrated performance improvements over unimodal baselines and standard concatenation techniques across several medical imaging datasets [8].

2.5.4 Limitations and Thesis Positioning

Despite these methodological developments, most studies evaluate multimodal fusion within the context of a single dataset. To better understand the robustness of these fusion strategies, it is valuable to assess their implementation across distinct data environments.

This thesis focuses on evaluating feature-fusion techniques for breast MC classification. This work assesses intermediate fusion (DAFT) against tabular-only and image-only baselines, as well as decision-level fusion, independently within two separate pipelines:

1. A pipeline utilizing 2D mammography combined with tabular demographic data.
2. A pipeline utilizing 3D Micro-CT combined with radiomic features.

By applying these fusion architectures to both settings, this thesis attempts to evaluate how feature-level and decision-level integration function across fundamentally different dimensionalities, data inputs, and classification tasks.

2.6 Advanced Feature-Level Conditioning: DAFT

To resolve the parameter imbalances and spatial disconnect of simple intermediate concatenation, researchers have developed advanced conditioning modules. These mechanisms are rooted in Feature-wise Linear Modulation (FiLM), an architecture originally designed for visual reasoning tasks that applies affine transformations to CNN feature maps based on external language inputs [34].

In the medical domain, this concept was adapted into DAFT by Wolf et al. [8]. The DAFT module dynamically conditions 2D or 3D convolutional feature maps using tabular context vectors, enabling a tightly coupled, bidirectional information exchange.

Mathematically, given an intermediate image feature map $X \in \mathbb{R}^{C \times D \times H \times W}$ (where C is the number of channels, and D , H , and W represent depth, height, and width) and a tabular feature vector $T \in \mathbb{R}^M$ (where M is the number of clinical variables), DAFT utilizes an auxiliary bottleneck network to map these inputs into scale (α) and shift (β) vectors [8]. Based on the formal definition by Wolf et al., the intermediate bottleneck vector z is generated by:

$$z = \text{ReLU}(W_1[\text{GAP}(X) \parallel T]) \quad (2.1)$$

The scaling and shifting vectors are then produced via secondary expansion layers. To maintain stability, these layers do not utilize bias terms:

$$\alpha = \sigma(W_\alpha z), \quad \beta = W_\beta z \quad (2.2)$$

The definitions of the equations' symbols are as follows:

- $\text{GAP}(X)$: Global Average Pooling applied across the spatial dimensions of the image feature map, yielding a vector of length C .
- $\parallel$: The concatenation operator, combining the pooled image vector and the tabular vector T .

- W_1 : The learnable weight matrix of the first dense layer that squeezes the concatenated vector into a bottleneck.
- W_α, W_β : The learnable weight matrices of the secondary dense layers that expand the bottleneck into length C .
- ReLU: The Rectified Linear Unit non-linear activation function.
- σ : An optional non-linear activation function (such as linear, sigmoid, or tanh) applied to restrict the scaling factor to a particular domain.
- z : The intermediate bottleneck activation vector.

Once α and β are generated, the original feature map X is affinely transformed across its spatial channels [8]:

$$\text{DAFT}(X, T) = \alpha \odot X + \beta \quad (2.3)$$

Here, $\odot$ denotes element-wise multiplication. By replacing standard concatenation with affine modulation, DAFT allows tabular data to actively scale (via α) and shift (via β) the spatial feature extraction of the CNN.

2.6.1 DAFT in Clinical Literature and Performance

In their original study, Wolf et al. evaluated DAFT on the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset, fusing 3D structural MRI scans with tabular clinical data (demographics and cognitive scores) [8]. They evaluated the architecture on two clinical tasks: Alzheimer’s Disease (AD) diagnosis (a classification task) and time-to-dementia prediction (a survival analysis task) [8].

The results demonstrated that DAFT outperformed both unimodal baselines and traditional fusion techniques [8]. For AD diagnosis, DAFT achieved a mean balanced accuracy of 0.622, surpassing standard late fusion (0.560) and intermediate concatenation strategies (ranging from 0.576 to 0.591) [8]. For the time-to-dementia survival task, the architecture reached a mean concordance index (C-index) of 0.748, showing a robust predictive performance by dynamically conditioning 3D spatial features with tabular variables [8].

With that in mind, the application of DAFT in the context of breast cancer—specifically for fusing 2D mammography/3D Micro-CT MCs with tabular data—remains unexplored, according to our research at the time of writing. This gap in the literature forms the primary motivation for the experimental pipelines evaluated in this thesis.

2.7 The Data-Efficiency Challenge in Medical Imaging

DL architectures typically require large, densely annotated datasets to generalize effectively without overfitting. In the medical domain, however, acquiring such datasets is inherently resource-intensive due to strict patient privacy constraints and the necessity for expert radiological consensus.

To help overcome this limitation, integrating tabular features—such as radiomics or clinical demographics—with image data has been demonstrated to improve model data efficiency. In a study evaluating the fusion of handcrafted features with CNNs, Gonidakis et al. demonstrated that joint-training multimodal architectures can achieve comparable or superior classification accuracy while requiring 33% to 43% less training data than unimodal image baselines [9].

However, effectively identifying and selecting the most impactful clinical or radiomic features remains a highly domain-specific challenge. The overall success of these multimodal architectures ultimately depends on isolating the tabular variables that provide the highest complementary value to the visual data [9].

Chapter 3

Methodology and Experimental Setup

The overall objective of this study is to systematically evaluate and compare distinct unimodal and multimodal data fusion strategies for the classification of breast MCs, thereby identifying the optimal architecture for this task.

This study was conducted across two distinct experimental pipelines to evaluate 2D clinical mammography and 3D high-resolution MC modality (Micro-CT).

3.1 Datasets

This study utilizes two distinct datasets to evaluate the proposed predictive pipelines. The first is a publicly available 2D mammography dataset (the EMBED dataset) combined with clinical demographic data [12]. The second is a highly curated, private 3D Micro-CT dataset of individual MCs combined with radiomic features [18].

3.1.1 The EMBED Dataset (2D Mammography)

For the initial phases of this study, we utilized the open-source partition of the EMBED dataset, which is promoted as a massive, diverse screening dataset representing 20% of the full proprietary cohort.

To ensure the networks were trained on radiologically valid data, a preprocessing pipeline was constructed to link the raw DICOM images with their corresponding patient metadata, Region of Interest (ROI) coordinates, and biopsy-proven ground truth labels (pathological severity). To maximize the utility of the severely limited data volume, the objective was structured as a binary classification task. The original multi-class pathological severities were dichotomized: severities 0, 1, 2, and 3 were grouped together as *actionable* findings, while severities 4 and 5 were designated as *non-actionable*. Representative examples of the full-field 2D mammograms from this dataset are shown in Figure 3.1.

However, despite the theoretical promise of a large-scale dataset, the practical utility of the open-source EMBED partition was severely compromised by rampant missing metadata.

The initial dataset contained 546,525 rows of tabular data and related 2D mammography images, each having 131 features, corresponding to 23,253 unique patients. After merging patient and accession IDs and filtering for laterality matches, 460,162 rows remained. Because the

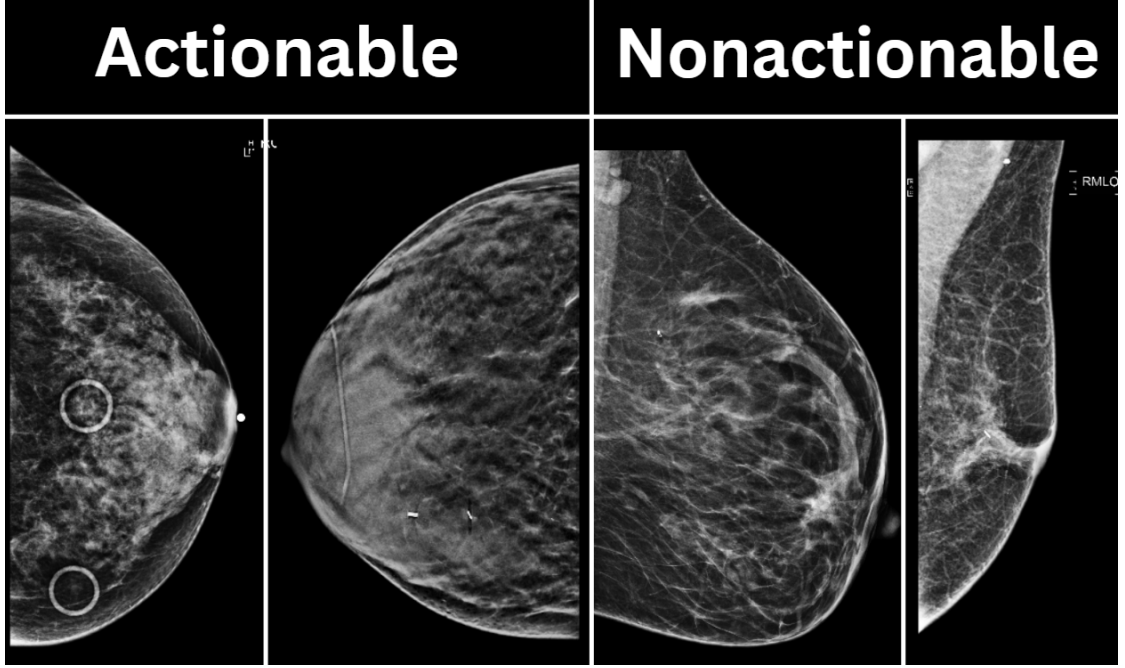


Figure 3.1: Representative full-field 2D mammograms from the EMBED dataset. The images are categorized into actionable (left) and non-actionable (right) cases based on their pathological severity. These images illustrate the varying breast structures and the different image sizes prior to ROI extraction.

EMBED dataset is a broad screening and diagnostic repository, the vast majority of these images represent either normal cases or abnormalities (such as masses or architectural distortions) that do not contain MCs. To construct a cohort specifically for MC analysis with valid ROIs and ground truth labels, the data was sequentially filtered:

- **Absence of Calcifications:** 429,799 rows were excluded because they represented normal findings or non-calcified lesions, leaving 30,363 rows that explicitly indicated the presence of MCs.
- **Missing ROI Coordinates:** 27,109 rows were excluded due to the absence of bounding box coordinate arrays, reducing the pool to 3,254 images with locatable MCs.
- **Missing Ground Truth:** A further 1,977 rows were excluded due to the lack of pathological severity labels required for supervised learning.

This strict isolation of MCs yielded a final dataset of only 1,277 image-to-finding pairs across just 455 unique patient visits (711 actionable, 566 non-actionable).

In an attempt to use a larger training cohort, the inclusion criteria were expanded to encompass *all* lesion types, rather than strictly MCs. Even under these relaxed parameters, dropping empty ROIs (-447,499 rows) and missing ground truth labels (-10,191 rows) yielded a maximum of only 2,472 image-to-finding pairs across 864 patient visits (1,303 actionable, 1,169 non-actionable).

Ultimately, the severe lack of paired ROI coordinates and pathological ground truths in the open-source EMBED dataset highlights a significant limitation in its current state for supervised

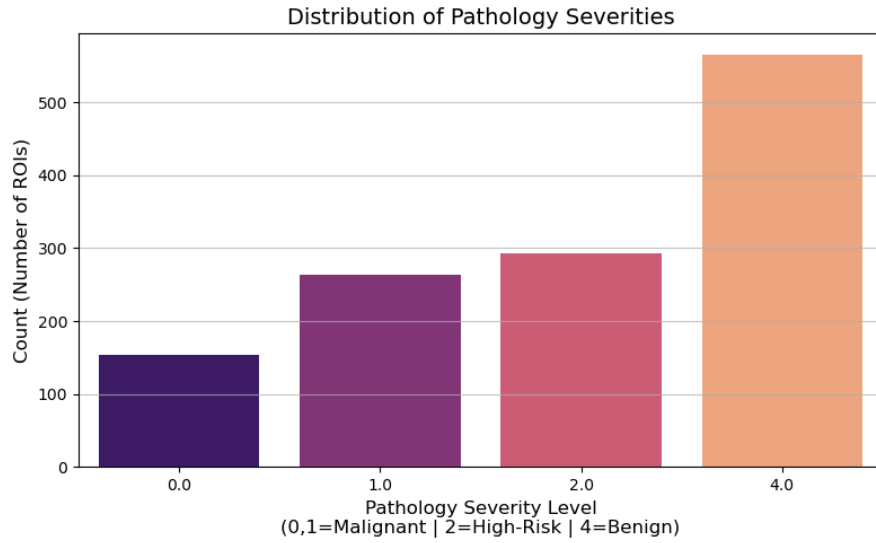


Figure 3.2: Distribution of pathological severities within the MCs-only EMBED dataset partition. This histogram highlights the frequency of each severity score (0–5) and illustrates the class imbalance that motivated the dichotomization of the data into actionable (0–3) and non-actionable (4–5) binary cohorts to preserve statistical viability during training.

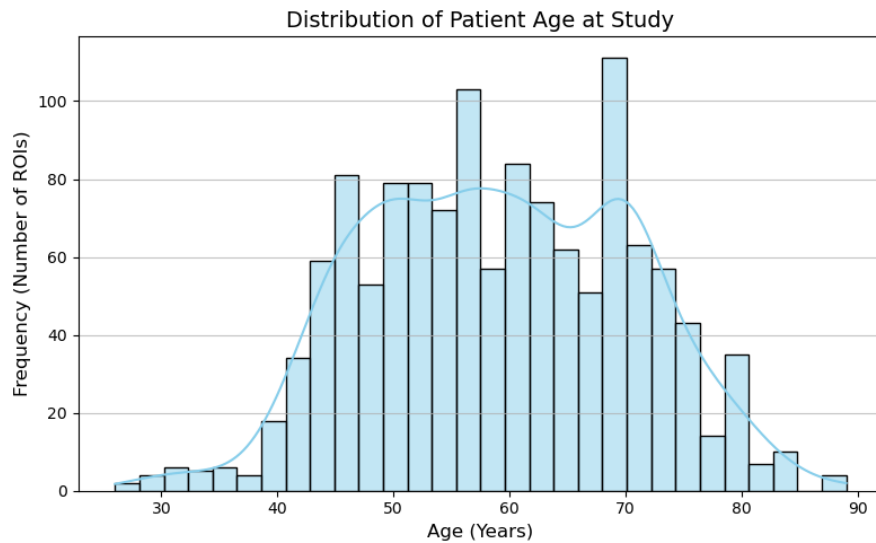


Figure 3.3: Age distribution of the patient cohort within the open-source MCs-only EMBED dataset partition. The histogram illustrates the demographic spread of the screening population included in the initial 2D mammography baseline experiments.

lesion-specific DL. The resulting data volume was deemed insufficiently small for training highly robust, generalizable multimodal networks.

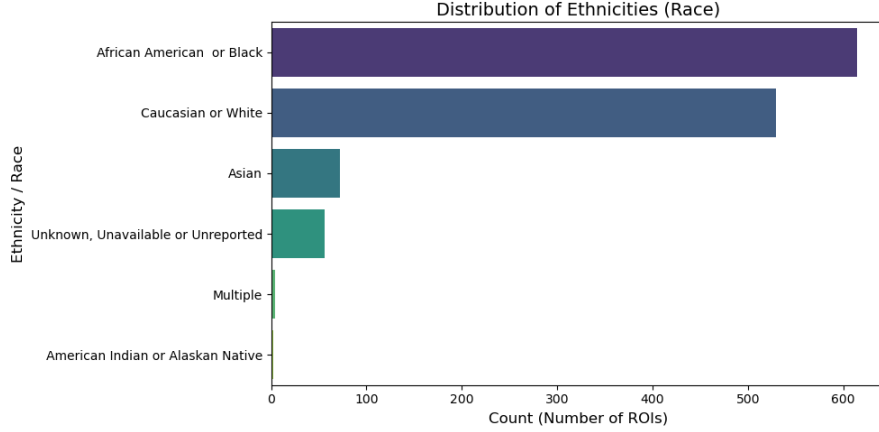


Figure 3.4: Distribution of self-reported race and ethnicity within the open-source MCs-only EMBED dataset partition. Visualizing this demographic composition provides context regarding the population-level diversity of the initial 2D mammography baseline models.

3.1.2 Private 3D Microcalcification Dataset

Due to the data sparsity encountered in the EMBED dataset, the primary multimodal experiments — specifically the intermediate DAFT fusion — were conducted using a proprietary dataset of 3D Micro-CT scans.

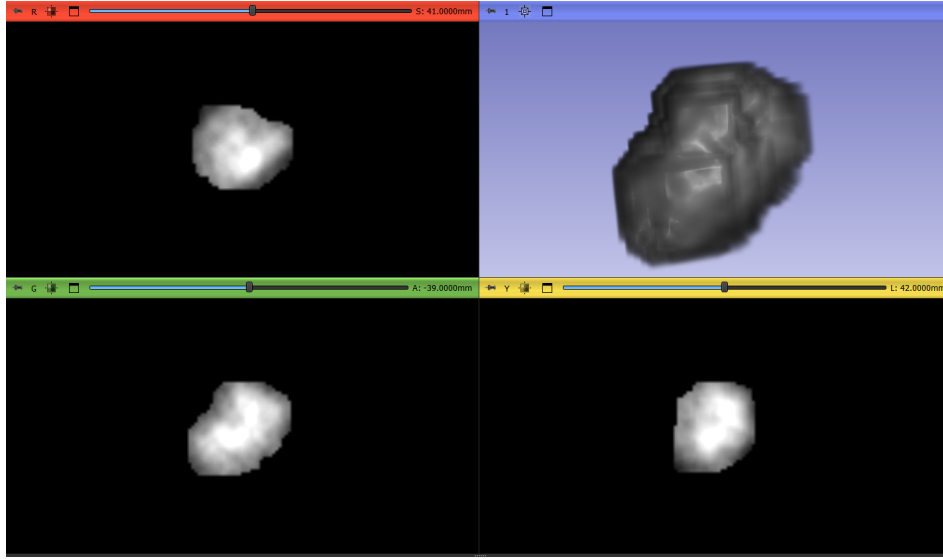
This curated dataset comprises 3,504 individual 3D MC images. Each of the 3,504 individual 3D image volumes is explicitly paired with a comprehensive vector of radiomic features and ground truth labels.

This complete, one-to-one mapping between the high-resolution 3D visual space and the tabular radiomic space provided a data-rich environment to attempt to train and evaluate the Late Fusion and Intermediate DAFT multimodal architectures.

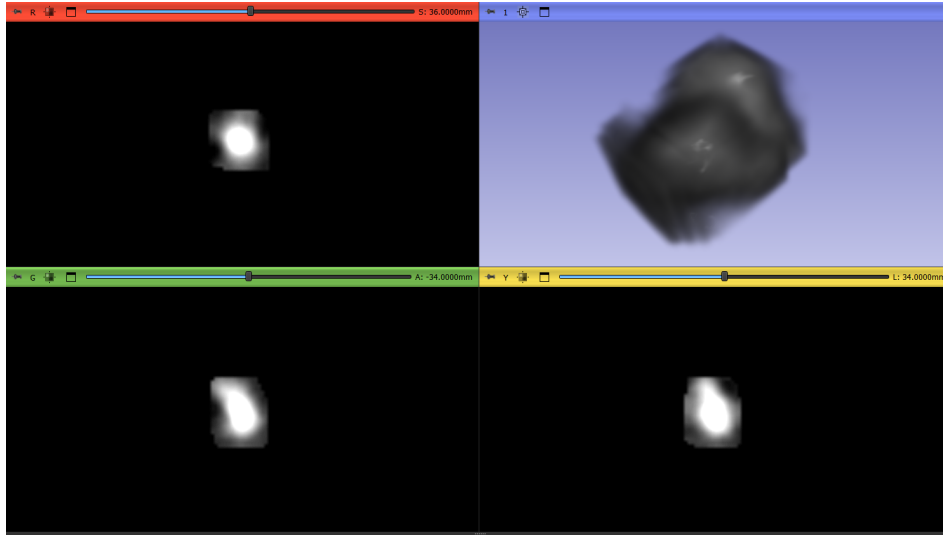
For radiomic features, radiomic feature extraction requires both the grayscale image and a binary mask defining the ROI. The MCs were first segmented using a previously validated threshold-based method, with a fixed threshold of 70 on the 8-bit intensity scale, followed by connected-component analysis [18].

However, recent work showed that the tissue surrounding the MCs also provides relevant diagnostic information [25]. This recent study conducted a sensitivity analysis to determine a specific dilation size that captures the most relevant diagnostic features [25]. Consequently, for this thesis, we used images obtained after dilation of the original mask with a cubic structuring element of size $(10 \times 10 \times 10)$. The dilated mask was multiplied voxel-wise with the original Micro-CT image, retaining the grayscale values within the dilated region while setting all voxels outside it to zero (to minimize background influence from entering the cropped ROI).

Radiomic features were then extracted from these images [18]. Classification experiments were performed using the complete radiomic feature set and different feature subsets obtained through recursive feature elimination. The subset of 200 features achieved the highest validation performance under the leave-one-subject-out evaluation framework, and these 200 features were therefore used in the subsequent analyses [25]. The top 200 features mentioned are used in the ablation study done in this thesis.



(a) Benign Microcalcification



(b) Malignant Microcalcification

Figure 3.5: Multi-planar and 3D volumetric samples of individual MCs from the private Micro-CT dataset, visualized using 3D Slicer. Each four-panel grid displays the axial, sagittal, and coronal 2D cross-sections alongside a corresponding 3D ray-cast surface render. (a) A benign MC morphology. (b) A malignant MC morphology.

3.1.3 EMBED Tabular and Image Preprocessing

Prior to spatial image extraction, the accompanying clinical metadata required systematic selection. The original EMBED dataset contained 131 unique clinical, technical, and administrative columns. To construct a tabular dataset suitable for the machine learning baselines and multi-

modal integration, variables were evaluated and excluded based on clinical relevance, usability (e.g., dropping columns with predominantly null values), and the risk of data leakage. Features that directly implied the ground truth diagnosis — such as further tests done only on higher severity cases — were removed in the preprocessing to prevent said data leakage. While BI-RADS is commonly utilized as a tabular feature in related studies, excluding BI-RADS was a necessary methodological choice for two primary reasons. First, the majority of the available cases were assigned a BI-RADS 0 assessment. According to the National Cancer Institute, a BI-RADS 0 score identifies an incomplete mammogram study requiring further evaluation, such as additional mammography views or an ultrasound, before a final assessment can be made [29]. Because this score lacks definitive diagnostic information, it cannot serve as a ground truth label, which is why pathological severity was utilized as the primary target variable instead. Second, for cases with definitive non-zero BI-RADS scores, including the assessment as an input feature would constitute direct data leakage, as it could provide the model with the radiologist’s final diagnostic conclusion.

This selection process isolated 20 essential variables. To explicitly prevent the appearance of data leakage, Table 3.1 separates the structural metadata and target labels from the actual predictive features used as inputs for the classification algorithms.

Table 3.1: Selected tabular variables from the EMBED dataset after filtering for clinical relevance, usability, and data leakage. To clarify data boundaries, the table explicitly separates metadata and labels (used only for cohort filtering and evaluation) from the features actually utilized as predictive inputs for the machine learning models.

Category	Variable Name	Description
Metadata & Ground Truth (Excluded from Predictive Input)		
Identifiers & Labels	empi_anon	Anonymized patient identifier
	acc_anon	Anonymized accession (study) identifier
	path.severity	Pathological severity score (original ground truth)
	target_label	Dichotomized binary target (Actionable vs. Non-actionable)
Imaging Metadata	anon_dicom_path	File path to the anonymized DICOM image
	ROI_coords	Spatial bounding box coordinates for the ROI
	num_ROI	Number of annotated ROIs in the image
	ImageLateralityFinal	Confirmed image laterality
Predictive Features (Utilized as Model Input)		
Clinical & Demographics	age_at_study	Patient age at the time of the imaging study
	ETHNIC_GROUP_DESC	Self-reported ethnic group (e.g., Hispanic/Latino)
	ETHNICITY_DESC	Self-reported race/ethnicity description
	tissueden	Categorical breast tissue density
Lesion Characteristics	calcfind	Descriptor of MC morphology
	calcdistri	Descriptor of MC spatial distribution
	size	Recorded size of the lesion
	location	Clock-face location of the finding in the breast
	depth	Relative depth of the finding
	distance	Distance of the finding from the nipple
	side	Indicated side of the breast (Left/Right)
	ViewPosition	Mammographic view angle (e.g., CC, MLO)

A complete list of all clinical, metadata, and DICOM features originally available in the

EMBED dataset prior to this selection process is provided in Appendix A.

Following tabular feature selection, the 2D DICOM mammography images required spatial and radiometric preprocessing prior to model ingestion. To ensure uniform input dimensions for the CNNs while preserving the dynamic range of the original scans, a fixed-window cropping algorithm was developed and used.

First, the pixel arrays were extracted from the DICOM files. Because mammography vendors utilize varying photometric interpretations, the algorithm systematically identified images encoded as `MONOCHROME1` (where dense tissue appears dark) and inverted their pixel intensities. This standardizes the intensity direction across the dataset, to make MCs appear as hyperintense (bright) signals against a darker background.

Next, the bounding box coordinates provided in the dataset metadata were utilized to compute the spatial center of each identified lesion. To construct a uniform tensor, a fixed spatial window of 512×512 pixels was dynamically generated around this calculated center point. To prevent data loss in edge cases where the lesion was located near the periphery of the breast, the algorithm automatically shifted the bounding window inward to remain strictly within the image dimensions. In rare instances where the native mammogram itself was smaller than the target window, the image was zero-padded (with black pixels) to enforce the 512×512 requirement.

Finally, to prepare the extracted spatial crops for neural network ingestion, the pixel arrays underwent Min-Max normalization, mapping the intensity values to a standardized 16-bit unsigned integer range (0 to 65535). Representative examples of these extracted ROI crops for both the actionable and non-actionable cohorts are visualized in Figure 3.6.

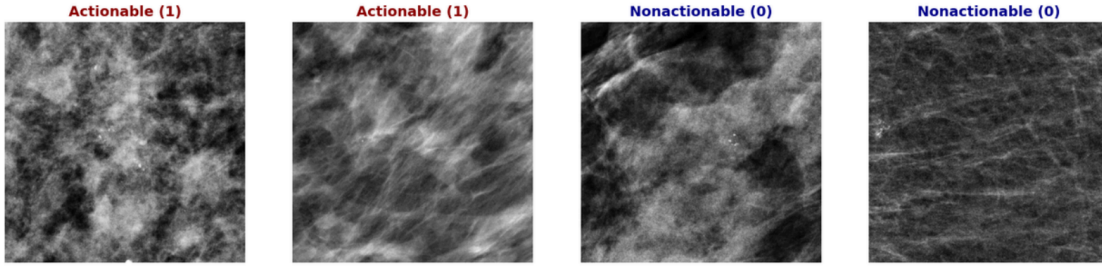


Figure 3.6: Representative examples of extracted ROI patches from the EMBED dataset. The panels display cropped mammographic images containing MCs. The patches are categorized according to the established binary classification threshold: Actionable findings (left, representing grouped severities 0–3) and Non-actionable findings (right, representing grouped severities 4–5). These spatial crops were ingested directly by the unimodal baseline architectures without additional contrast enhancement.

During the initial development of the image-based pipelines, Contrast-Limited Adaptive Histogram Equalization (CLAHE) was evaluated as a potential preprocessing step. In classical medical image analysis, CLAHE is frequently applied to mammograms to enhance the local conspicuity of subtle findings, such as MCs, against dense fibro-glandular background tissue.

However, tests across the baseline CNNs revealed that the application of CLAHE consistently reduced the performance of the models. Consequently, CLAHE was not used for the final pipelines. The neural networks were instead trained directly on the cropped and normalized ROIs of the mammography images without any additional contrast enhancement.

3.1.4 3D Micro-CT Image Preprocessing and Spatial Normalization

Prior to ingestion by the 3D CNNs in the second experimental pipeline, the segmented Micro-CT volumes required radiometric and spatial standardization.

First, the native voxel arrays were cast to 32-bit floating-point representations. The original Micro-CT volumes natively occupied an 8-bit intensity space (0 to 255), with maximum intensities varying based on the specific composition of each MC. A Min-Max normalization was applied, linearly scaling the voxel intensities to a continuous range between 0.0 and 1.0.

Following radiometric normalization, spatial standardization was performed. DL architectures require fixed-size input tensors; however, the extracted MCs naturally varied in physical dimensions. To standardize the tensor size while minimizing artificial stretching, an aspect-ratio-preserving resizing algorithm was implemented.

For each volume, the longest spatial dimension (across the X, Y, or Z axes) was identified. A uniform scaling factor was computed to resize this maximum dimension to 64. The entire 3D volume was then scaled by this singular factor. This approach was chosen to maintain the natural macroscopic proportions of the MCs as much as possible, though the resampling and interpolation process inevitably modifies the underlying voxel representation to some degree.

Because the aspect ratio was maintained, the newly scaled volumes were typically smaller than 64 voxels in the remaining two dimensions. To achieve the desired $64 \times 64 \times 64$ uniform tensor, symmetrical zero-padding was dynamically calculated and applied to the deficient axes. This padding effectively centered the MC within a uniform $64 \times 64 \times 64$ empty background. This ensured that all 3D images fed into the CNNs and fusion networks possessed identical tensor dimensions and a standardized intensity range (0.0 to 1.0), without disproportionately distorting the structural morphology of the lesions.

3.1.5 Cross-Validation Strategy and Patient-Level Splitting

For the EMBED dataset, model evaluation was conducted using a standard 5-fold CV at the patient level with class balance stratification. Due to the limited amount of usable data in this subset, the choice was made to use only a training-test split instead of a training-validation-test split. This limitation should be taken into consideration when interpreting the generalizability of the results in the first pipeline.

For the private dataset, the proprietary 3D dataset was also partitioned at the patient level to prevent data leakage. Because multiple individual MCs can be extracted from a single patient’s biopsy volume, standard randomization at the image level risks placing highly correlated lesions from the same biological environment into both the training and test sets.

To counteract this, a patient-wise 10-fold CV scheme was implemented. The data was grouped by unique patient identifiers and stratified by their severity label (Benign vs. Malignant). Across the entire dataset, a total of 94 unique patients were identified.

For each of the 10 folds, 10 benign and 10 malignant patients were sampled using fixed random seeding. These 20 patients were evenly divided to form balanced validation and test cohorts (5 benign and 5 malignant patients per split). All remaining unselected patients were allocated to the training cohort. Furthermore, the generation logic ensured that no two folds utilized the exact same patient distribution.

This controlled protocol ensured that across all 10 independent folds, the models were trained on 74 patients, validated on 10 patients, and tested on 10 patients, providing a standardized evaluation framework that prevented patient-level data leakage.

3.2 Overview of Experimental Pipelines

To systematically evaluate the diagnostic added value of multimodal fusion models, the experimental pipelines were structured to start with a Tabular-only and Image-only baselines, and progress to using the more complex fusion models. Figure 3.7 visualizes this phased, sequential approach.

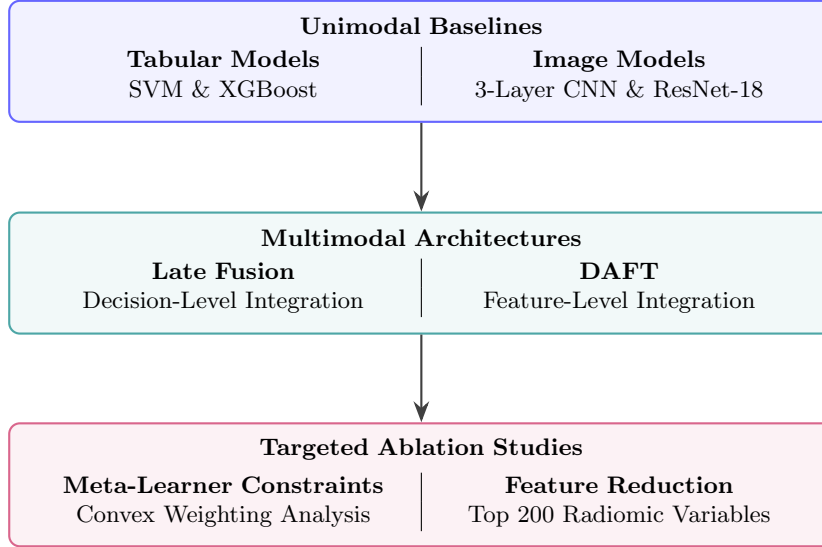


Figure 3.7: A visual schema of the experimental pipelines. The methodology follows a linear progression from evaluating independent unimodal baselines (blue), to implementing multimodal fusion strategies (teal), and concluding with targeted ablation studies (purple). Each phase is divided into its two parallel experimental components.

The predictive models were developed, optimized, and evaluated in the following sequential order:

1. **Unimodal Tabular Baselines:** Traditional machine learning classifiers, specifically SVM and XGBoost, were trained exclusively on tabular data from the EMBED dataset in the first pipeline, and on the handcrafted radiomic feature vectors from the private dataset in the second pipeline.
2. **Unimodal Image Baselines:** 2D/3D DL architectures, including a custom 3-layer CNN and a Standard/3D ResNet-18, were trained exclusively on the standardized Micro-CT volumetric images.
3. **Late Fusion (Decision-Level):** A stacking ensemble approach was implemented utilizing the best-performing configurations from the tabular and image baselines. The independent predictive probabilities generated by these unimodal networks were concatenated and passed into a Logistic Regression meta-classifier to derive the final multimodal prediction.
4. **Intermediate Fusion (Feature-Level):** The DAFT architecture was trained end-to-end. This approach actively conditioned the intermediate image feature maps using the tabular data.

5. **Targeted Ablation Studies:** To investigate modality dominance and feature redundancy, supplementary ablation studies were conducted. These included constraining the Late Fusion meta-learner weights to analyze positive-additive voting dynamics, and performing feature elimination (restricting the tabular input to the Top 200 radiomic features) to possibly denoise the intermediate DAFT bottleneck.

By establishing unimodal baselines first, this phased approach attempts to show that any performance gains observed in the late and intermediate fusion models can be quantified and attributed to the multimodal integration strategies.

3.3 Late Fusion Architecture (Decision-Level Integration)

To establish a multimodal baseline, a late fusion—or decision-level integration—approach was implemented. Unlike intermediate fusion architectures that combine data streams during the feature extraction process, late fusion preserves the independence of the unimodal pipelines until the final predictive stage.

The rationale behind this architecture is to allow each modality-specific network to learn its decision boundary without cross-modal interference. By separating the high-dimensional spatial data (2D mammography and 3D Micro-CT) from the lower-dimensional structured data (tabular data), the unimodal networks mitigate the risk of early feature dominance, where a joint model might disproportionately weight one modality over another during the initial training phases.

3.3.1 Mathematical Formulation and Aggregation

The Late Fusion pipeline consists of three distinct phases: independent model optimization, probability extraction, and meta-classification.

Let X_{img} denote the 3D Micro-CT image volume and X_{tab} denote the corresponding tabular radiomic feature vector. First, the unimodal image model (3D ResNet-18) and tabular model (XGBoost) are trained independently using their respective optimized hyperparameters and early stopping criteria.

During the inference phase, these optimized models function as independent predictors that map their respective inputs to a probability space. Let $\hat{y}_{img}$ and $\hat{y}_{tab}$ represent the predicted probabilities of malignancy generated by the image network $f_{img}(\cdot)$ and the tabular network $f_{tab}(\cdot)$:

$$\hat{y}_{img} = f_{img}(X_{img}), \quad \hat{y}_{tab} = f_{tab}(X_{tab}) \quad (3.1)$$

Once these independent uni-modal decisions are extracted, they are aggregated to form the final predictive output. While standard aggregation layers often utilize simple averaging or hard voting mechanisms, this study employs a learned meta-model approach.

The individual probabilities are concatenated into a new, two-dimensional feature vector $V_{meta} = [\hat{y}_{img}, \hat{y}_{tab}]$. This vector is fed into a Logistic Regression meta-classifier, $g(\cdot)$, which learns to weight the image prediction against the tabular prediction to output the final diagnostic probability, $\hat{y}_{final}$:

$$\hat{y}_{final} = g([\hat{y}_{img}, \hat{y}_{tab}]) \quad (3.2)$$

By utilizing a Logistic Regression meta-classifier at the aggregation layer, the Late Fusion pipeline accounts for the varying predictive strengths of the constituent unimodal models, yielding a combined final diagnosis.

Furthermore, to investigate the underlying voting dynamics of the meta-classifier, a targeted ablation study was conducted. To prevent the ensemble from assigning negative penalties to a

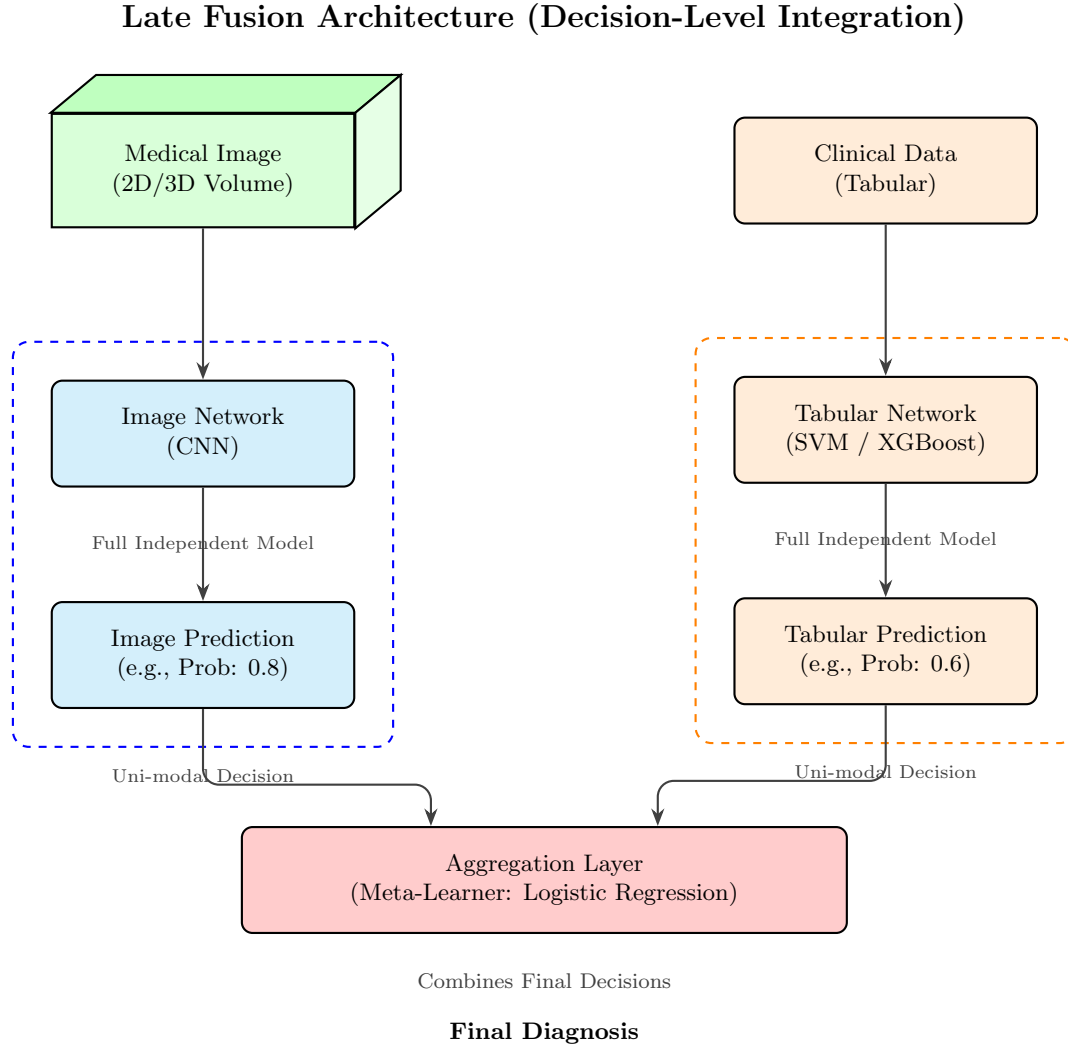


Figure 3.8: Overview of the Late Fusion (Decision-Level Integration) architecture. The multimodal pipeline processes the medical images and tabular clinical data through independent unimodal networks. The resulting uni-modal predictions are subsequently concatenated and passed into an aggregation layer (meta-classifier) to compute the final diagnosis.

weaker modality, a convex combination constraint was implemented. Specifically, the learned coefficients of the meta-learner were passed through a Softmax function, forcing the weights assigned to each modality to be strictly non-negative and sum to exactly 1.0. This formulation restricted the aggregation layer to act as a strict weighted average, enabling a direct and interpretable quantification of each modality’s contribution to the final prediction.

3.4 The Dynamic Affine Feature Map Transform (DAFT) Architecture

To fuse the spatial image tensors with the lower-dimensional tabular variables, this study implemented DAFT. As detailed in the literature review, DAFT integrates modalities by dynamically scaling and shifting the intermediate feature maps of a convolutional layer [8]. In both experimental pipelines, the DAFT module was placed within the final residual block of the respective ResNet-18 backbones.

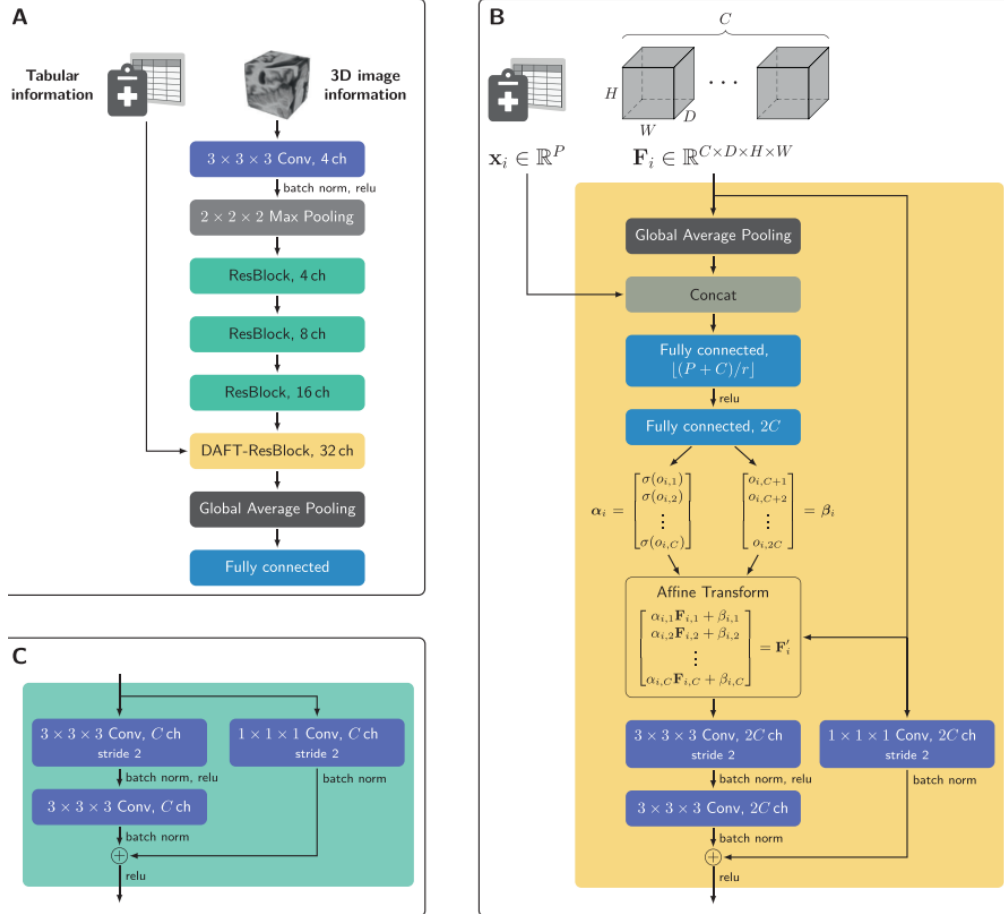


Figure 3.9: Overview of the DAFT architecture. The module squeezes spatial dimensions via global average pooling, concatenates the output with tabular information, and passes the combined vector through an auxiliary bottleneck network to generate scale and shift parameters to affinely transform the input feature maps. Taken from Wolf et al. [8].

3.4.1 Mathematical Formulation and Dimensional Adaptation

Let $x_i \in \mathbb{R}^P$ denote the tabular feature vector for the i -th patient, where P represents the number of tabular variables. Let $F_{i,c}$ denote the c -th channel of the intermediate image feature map for

that same patient [8]. Depending on the pipeline, the spatial dimensions of this feature map vary: it is $H \times W$ for the 2D mammography pipeline, and $D \times H \times W$ for the 3D Micro-CT pipeline.

The DAFT module transforms these feature maps by computing a channel-specific scale factor ($\alpha_{i,c}$) and shift offset ($\beta_{i,c}$):

$$F'_{i,c} = \alpha_{i,c}F_{i,c} + \beta_{i,c} \quad (3.3)$$

These affine parameters are generated by an auxiliary neural network that maps both the pooled image features and the tabular vector to the required scalar values:

$$\alpha_{i,c} = f_c(F_{i,c}, x_i), \quad \beta_{i,c} = g_c(F_{i,c}, x_i) \quad (3.4)$$

To compute α_i and β_i , the implemented auxiliary network executes the following sequence of operations based on the original formulation by Wolf et al. [8]:

1. **Global Average Pooling (GAP):** The spatial dimensions of the image feature map F_i are pooled to create a 1D vector. This pooling occurs across height and width for the 2D pipeline, and across depth, height, and width for the 3D pipeline.
2. **Concatenation:** The pooled image vector is concatenated with the tabular vector x_i .
3. **Bottleneck Compression:** The concatenated vector passes through a Fully Connected (FC) layer with a ReLU activation, squeezing the representation into a predefined bottleneck dimension.
4. **Parameter Expansion:** A secondary FC layer expands the bottleneck vector into the final $\alpha_i \in \mathbb{R}^C$ and $\beta_i \in \mathbb{R}^C$ output vectors. To maintain stability, these expansion layers do not utilize bias terms.

Extension: Dynamically Weighted DAFT (Ablation Study)

To investigate the relative contributions of the scaling and shifting operations during the 3D ablation study, an extended DAFT formulation was evaluated. This extension modifies the original DAFT architecture in two ways: it introduces global, learnable scalar weights to control the magnitude of the modulation, and it alters the mathematical formulation by introducing a residual component to the scaling operation. The modified transformation is defined as:

$$F'_{i,c} = (1 + \sigma(w_\alpha)\alpha_{i,c})F_{i,c} + (\sigma(w_\beta)\beta_{i,c}) \quad (3.5)$$

The scalar weights w_α and w_β are initialized at 0.0. By passing them through a sigmoid activation function, $\sigma(\cdot)$, their initial effective weighting evaluates to exactly 0.5. During training, the optimizer independently adjusts w_α and w_β to dynamically scale the magnitude of the affine modulation, testing if this can provide a soft gating mechanism to regulate the tabular data's overall influence on the spatial feature maps.

As the formulation demonstrates, the addition of the base feature map ($1 \cdot F_{i,c}$) structurally alters the original DAFT formulation, making this an architectural modification rather than a simple weighting mechanism. Furthermore, the scalar weights w_α and w_β strictly measure the global importance of the scaling and shifting operations across the network; they do not measure or indicate the predictive contribution of individual tabular variables.

3.5 Pipeline 1: 2D EMBED Dataset

The first pipeline evaluated an open-source subset of the EMBED dataset [12], consisting of 1,225 patients with 2D cropped mammography ROIs paired with the tabular features mentioned in the Datasets section and in Appendix A.

As established in the methodology, models were evaluated using a patient-level 5-fold CV. As mentioned in the datasets section, to maximize the utility of the limited data, each fold was partitioned strictly into a training and a test split, omitting an independent validation cohort. The evaluated architectures included:

- **Image Models:** ResNet-18 backbones trained on ROIs of the 2D mammography images.
- **Tabular Models:** Linear SVM and XGBoost models operating on preprocessed tabular clinical variables.
- **Multimodal Architectures:** Late Fusion (ResNet-18 + SVM), and DAFT (joint pooling and bottleneck concatenation) [8].

3.6 Pipeline 2: 3D Private Dataset (Micro-CT + Radiomics)

To evaluate multimodal fusion in a distinct, ex-vivo research setting, the second pipeline utilized the private 3D Micro-CT dataset previously mentioned. This proprietary dataset contained 3,504 individual MC volume patches ($64 \times 64 \times 64$ voxels) explicitly paired with 979 detailed PyRadiomics features across 94 patients [18]. The evaluated architectures and hyperparameter configurations included:

- **Tabular Models:** Custom 10-Fold Grid Search for SVM ($C \in \{0.1, 1, 10\}$, Kernels: Linear, RBF, Poly) and XGBoost (243 hyperparameter combinations covering learning rate, tree depth, and subsampling rates).
- **3D Vision Models:** Custom 10-Fold Grid Search for a simple 3-Layer 3D CNN and a 3D ResNet-18 trained from scratch using mixed-precision training.
- **3D Multimodal Models:** 3D Late Fusion (3D ResNet-18 + XGBoost + Logistic Regression Meta-Learner) and a Custom 10-Fold Grid Search for 3D DAFT operating on a ResNet (`r3d_18`) backbone.

3.7 Statistical Evaluation Methods

To evaluate the statistical significance of the observed performance differences between the competing unimodal and multimodal architectures, statistical testing was implemented. The predictions generated during the holdout test phases were aggregated for pairwise statistical comparisons. Because the 3D Micro-CT dataset contains multiple MCs per patient, the statistical evaluation was explicitly designed to account for intra-patient correlation rather than treating individual lesions as independent observations.

The ROC-AUC was evaluated utilizing a non-parametric patient-level clustered bootstrapping simulation. For each pairwise model comparison, patients—along with all of their associated MC predictions—were randomly resampled with replacement for 10,000 iterations to construct 95% Confidence Intervals (CIs) for the difference in AUC (ΔAUC). A performance difference was

considered statistically significant if the 95% CI did not include zero, which mathematically corresponds to a two-tailed p -value < 0.05 .

Furthermore, to evaluate the absolute error rates, binary classifications were obtained using a predefined probability threshold of $t = 0.5$. It is important to note a structural distinction in the evaluation levels: while the general classification metrics (Accuracy, F1-Score, Sensitivity, Specificity) reported in the results are computed at the individual MC level ($N = 3,504$), the statistical testing was conducted strictly at the patient level ($N = 94$). To account for intra-patient clustering and satisfy the assumption of independent observations required for significance testing, the individual MC probabilities for each patient were first averaged to derive a single aggregate diagnostic risk score per patient. The statistical significance of the disagreements in these patient-level binary predictions was then calculated using McNemar’s Test for paired nominal data. This provided context regarding whether the fusion models altered the absolute volume of correct patient diagnoses compared to radiomics in isolation.

Chapter 4

Results

In this chapter, we present the empirical results obtained from the evaluated unimodal and multimodal architectures. The structure of presenting these results is as follows: First, we report the classification performance and learning dynamics of the models evaluated in Pipeline 1, utilizing the 2D mammography EMBED dataset. Next, we present the primary results for Pipeline 2, detailing the performance of the architectures on the 3D Micro-CT and radiomics dataset. Following this, we present a targeted ablation study evaluating the impact of radiomic feature space reduction on the models used in this thesis. We then provide a statistical evaluation of the pairwise model comparisons to determine the statistical significance of the observed performance differences. Finally, we conclude the chapter with secondary ablation studies analyzing the voting dynamics of a constrained decision-level late fusion meta-learner, and giving the shift and scale variables in the DAFT network a learnable weight to determine each variable’s impact on the network.

4.1 Pipeline 1 Results (2D EMBED Dataset)

We summarize the test performance across the 5-fold CV evaluation on the MCs-only 2D EMBED dataset in Table 4.1.

Table 4.1: Test performance summary for Pipeline 1 (restricted to MCs in the 2D EMBED Dataset). All reported metrics represent the mean across the 5-fold cross-validation.

Modality	Architecture	Mean Test AUC	Accuracy	F1-Score	Sensitivity	Specificity
Image Only	Simple 3-Layer CNN	0.5738 ± 0.0310	0.5482	0.5261	0.5183	0.5781
Image Only	ResNet-18	0.5934 ± 0.0219	0.5614	0.5419	0.5342	0.5886
Tabular Only	SVM (Linear)	0.5957 ± 0.0577	0.5671	0.5523	0.5418	0.5924
Tabular Only	XGBoost	0.5635 ± 0.0560	0.5391	0.5184	0.5097	0.5685
Multimodal	Late Fusion	0.6378 ± 0.0238	0.6015	0.5892	0.5811	0.6219
Multimodal	DAFT (Intermediate Fusion)	0.6750 ± 0.0208	0.6483	0.6361	0.6294	0.6672

In this primary cohort, the unimodal image architectures (3-Layer CNN and ResNet-18) yielded mean test AUCs of 0.5738 and 0.5934, respectively. The tabular baselines, SVM and XGBoost, achieved mean test AUCs of 0.5957 and 0.5635. The Late Fusion ensemble obtained a mean test AUC of 0.6378. The feature-level intermediate fusion via the DAFT architecture

yielded the highest AUC in this setting, achieving a mean test AUC of 0.6750. The best-performing configuration for this architecture utilized a learning rate of 0.001, a bottleneck dimension of 64, and a dropout rate of 0.1, selected based on the CV test folds.

To visualize the aggregated predictive capability of the intermediate fusion approach across the dataset, the mean ROC curve across all five folds for the evaluated DAFT configuration is presented in Figure 4.1.

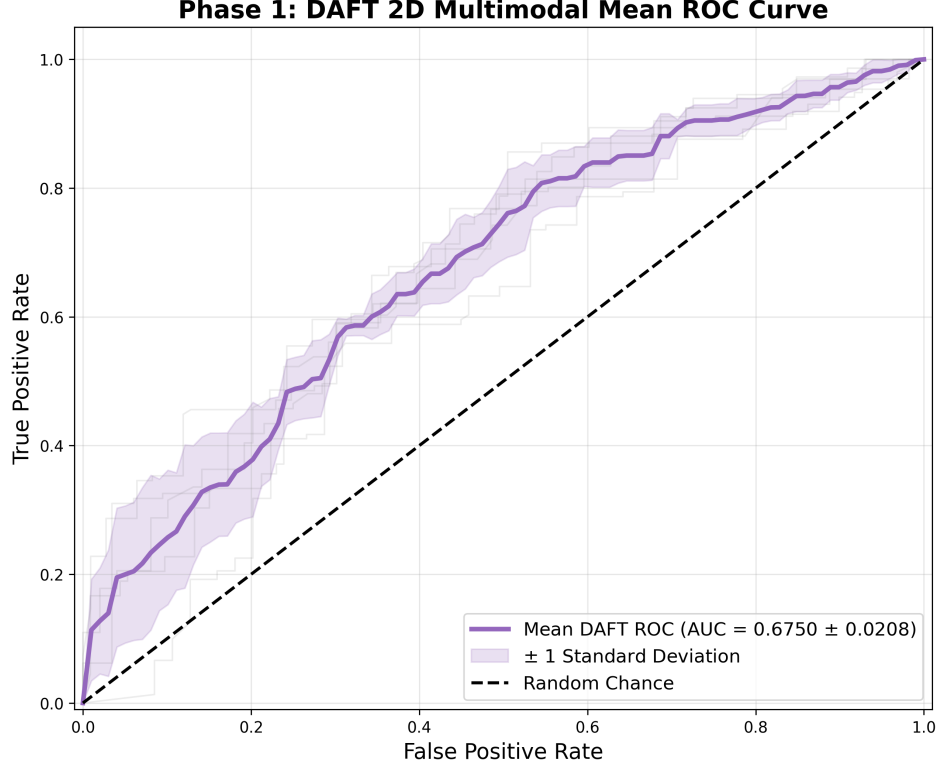


Figure 4.1: Mean ROC curve across 5-fold CV for the top-performing DAFT configuration on the 2D EMBED dataset. The shaded region represents ± 1 standard deviation across the test folds.

Sensitivity Analysis: Expanded Lesion Cohort

To assess whether model performance was constrained strictly by the limited sample size of the MCs-only cohort, a sensitivity analysis was conducted utilizing an altered experimental setting. The dataset composition was expanded to include all available lesion types with valid ground truth labels, not only those containing MCs, increasing the available data to 2,472 image-to-finding pairs. The tabular feature indicating whether an image contained MCs or not was added back to the tabular features used for this sensitivity analysis. The models were evaluated using the identical 5-fold CV protocol. The comparative test results across the predictive pipelines for this expanded cohort are summarized in Table 4.2.

In this expanded dataset, performance across the tabular and multimodal pipelines generally did not show a noticeable improvement compared to the MCs-only subset. The unimodal XG-

Table 4.2: Test performance across all predictive pipelines using the expanded EMBED dataset (inclusive of all lesion types). All reported metrics represent the mean across the 5-fold cross-validation

Modality	Architecture	Mean Test AUC	Accuracy	F1-Score	Sensitivity	Specificity
Image Only	Simple 3-Layer CNN	0.5693 ± 0.0341	0.5431	0.5218	0.5142	0.5721
Image Only	ResNet-18	0.5699 ± 0.0219	0.5487	0.5304	0.5219	0.5756
Tabular Only	SVM (Linear)	0.6467 ± 0.0252	0.6124	0.5981	0.5823	0.6426
Tabular Only	XGBoost	0.6569 ± 0.0239	0.6238	0.6117	0.5984	0.6491
Multimodal	Late Fusion	0.6570 ± 0.0238	0.6251	0.6153	0.6072	0.6429
Multimodal	DAFT (Intermediate Fusion)	0.6718 ± 0.0158	0.6412	0.6378	0.6285	0.6539

Boost model achieved a mean test AUC of 0.6569, while the decision-level Late Fusion approach yielded a mean test AUC of 0.6570. The DAFT architecture again yielded the highest overall performance, reaching a mean test AUC of 0.6718. The best-performing configuration for this architecture utilized a learning rate of 0.0001, a bottleneck dimension of 32, and a dropout rate of 0.2.

4.2 Pipeline 2 Results (3D Micro-CT Individual MCs + Radiomics)

Table 4.3 presents the performance metrics for the 3D Micro-CT dataset evaluated using a patient-level 10-fold CV. For the DL architectures, early stopping was applied based on the validation loss to reduce overfitting, and the final reported metrics are the means across the 10 independent test folds.

Table 4.3: Test performance comparison for Pipeline 2 utilizing the full radiomic feature set. Metrics represent the mean across the 10-fold cross-validation. Model selection and holding are based on the highest Test AUC.

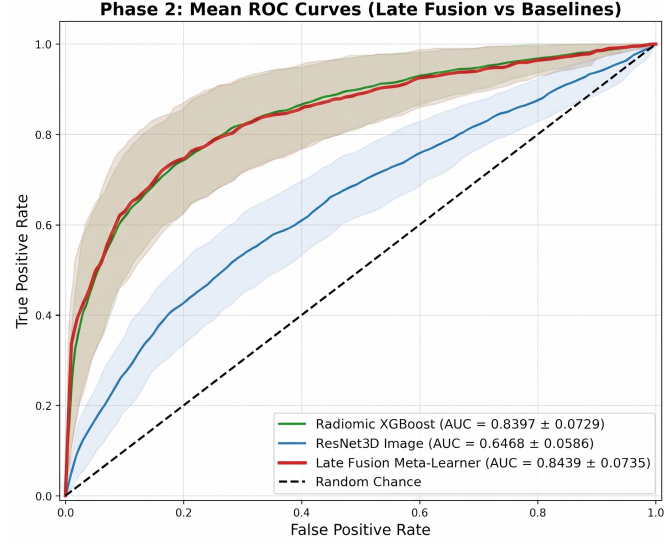
Modality	Architecture	Optimal Parameters	Test AUC	Accuracy	F1-Score	Sensitivity	Specificity
Tabular	SVM (RBF)	Kernel: RBF, $C = 1$, $\gamma = \text{scale}$	0.8109 ± 0.0520	0.7241	0.6721	0.7381	0.7201
Tabular	XGBoost	lr: 0.05, depth: 3, est: 200	0.8397 ± 0.0729	0.7485	0.6997	0.7651	0.7391
3D Image	Simple 3D CNN	lr: 1e-4, bs: 16, drop: 0.6	0.6617 ± 0.0610	0.5488	0.4117	0.4508	0.6674
3D Image	3D ResNet-18	lr: 1e-4, weight_decay: 1e-3	0.6468 ± 0.0586	0.5658	0.4291	0.4602	0.6796
Multimodal	3D Late Fusion	Meta: Balanced LogReg	0.8439 ± 0.0735	0.7390	0.7016	0.7962	0.7056
Multimodal	3D DAFT	lr: 1e-3, dim: 64, drop: 0.5	0.8062 ± 0.0641	0.7481	0.6718	0.7251	0.7583

In the unimodal evaluations, the deep vision architectures achieved a mean test AUC of approximately 0.65 to 0.66. The tabular model utilizing explicitly the radiomics features via XGBoost achieved a higher mean test AUC of 0.8397.

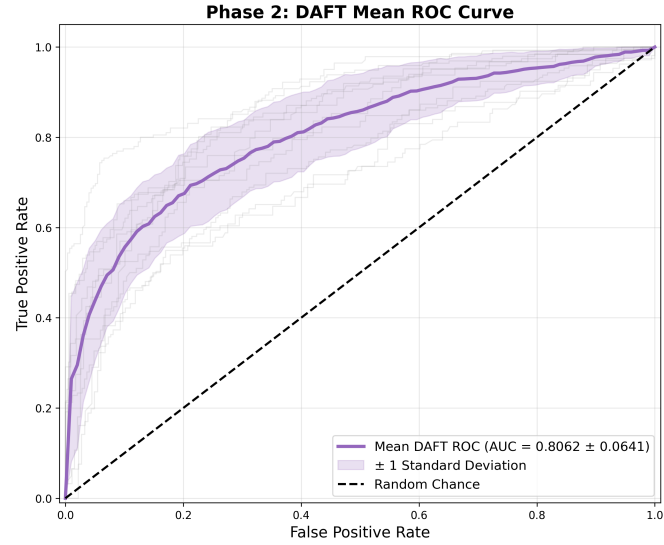
For the multimodal architectures, the intermediate fusion 3D DAFT model achieved a mean test AUC of 0.8062, representing a 0.1594 absolute AUC increase over the unimodal 3D ResNet-18 baseline. The 3D Late Fusion approach yielded a mean test AUC of 0.8439. Analysis of the Logistic Regression meta-learner for this configuration revealed a learned coefficient of +11.81 for the XGBoost probability and -2.70 for the 3D CNN probability.

To visualize the aggregated performance of these models, Figure 4.2 presents the mean ROC

curves across all 10 folds for the unimodal baselines and the Late Fusion meta-learner, as well as the independent mean ROC curve across all 10 folds for the DAFT architecture.



(a) Unimodal Baselines vs. Late Fusion



(b) Intermediate Fusion (DAFT)

Figure 4.2: Mean ROC curves across 10-fold CV on the test sets of the 3D Micro-CT dataset. (a) Compares the predictive ranking of the pure radiomic baseline (XGBoost), the pure visual baseline (3D ResNet-18), and the decision-level Late Fusion meta-learner. (b) Displays the predictive ranking of the feature-level intermediate fusion DAFT architecture. Shaded regions represent ± 1 standard deviation.

4.3 Ablation Study: Radiomic Feature Reduction (Top 200 Features)

In the primary 3D experiments, the models were trained utilizing the complete, unreduced set of radiomic features. However, high-dimensional radiomic datasets frequently exhibit feature redundancy and multi-collinearity, which can introduce statistical noise.

To evaluate the impact of feature space dimensionality on multimodal integration, an ablation study was conducted utilizing a reduced feature set. As detailed in the methodology, this specific subset of 200 features was obtained through recursive feature elimination in previous work on this dataset, having achieved the highest validation performance under a leave-one-subject-out evaluation framework [18]. By strictly restricting the tabular input space to these curated radiomic features, the unimodal baselines, Late Fusion ensemble, and DAFT architecture were retrained and evaluated under the identical patient-level 10-fold CV framework.

4.3.1 DAFT Hyperparameter Re-Optimization

Because the dimensionality of the tabular input was significantly reduced, the DAFT architecture required a secondary hyperparameter grid search to identify the optimal bottleneck configuration for the refined feature space. The results of this grid search are summarized in Table 4.4.

Table 4.4: 10-Fold CV test results for the DAFT ablation study using the reduced subset of the top 200 radiomic features.

Config (LR, Dim, Drop)	Mean Test AUC	Test Acc	Test F1
lr: 1e-3, dim: 64, drop: 0.5	0.8328 $\pm$ 0.0716	0.7568	0.6988
lr: 1e-4, dim: 64, drop: 0.2	0.8324 $\pm$ 0.1172	0.7471	0.6828
lr: 1e-3, dim: 32, drop: 0.5	0.8170 $\pm$ 0.1096	0.7345	0.6753
lr: 1e-3, dim: 64, drop: 0.2	0.8320 $\pm$ 0.0996	0.7423	0.6870

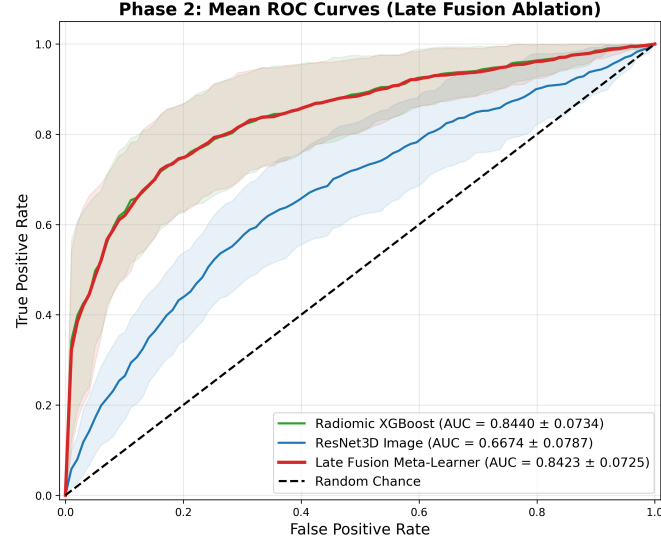
4.3.2 Ablation Results and Analysis

Following the optimization of the DAFT bottleneck, the performance of the multimodal pipelines utilizing the reduced feature subset was evaluated. To clearly present the effect of the feature reduction, Table 4.5 presents a direct side-by-side comparison of the Mean Test AUCs obtained using the full feature set versus the Top 200 subset.

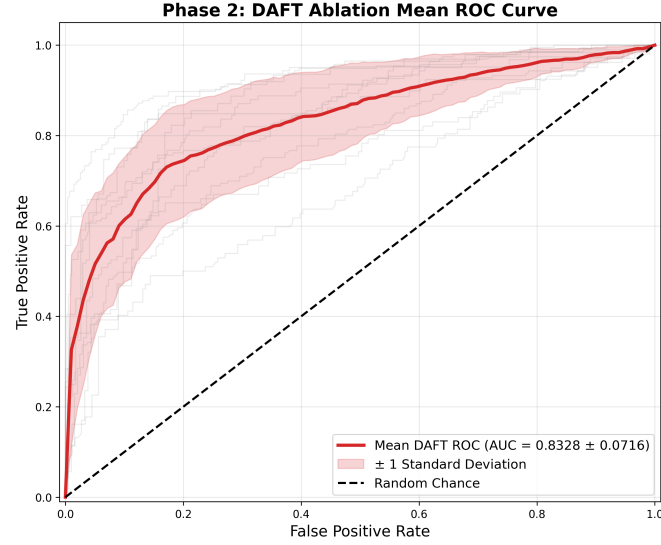
Table 4.5: Performance comparison demonstrating the effect of reducing the radiomic feature set to the top 200 features. The 3D ResNet-18 model does not utilize radiomic features and is included strictly as a visual reference baseline.

Modality	Model	Full Features Test AUC	Top 200 Features Test AUC	Top 200 Acc	Top 200 F1
Tabular	XGBoost	0.8440 $\pm$ 0.0734	0.8440 $\pm$ 0.0734	0.7527	0.7048
3D Image	3D ResNet-18*	0.6468 $\pm$ 0.0586	0.6674 $\pm$ 0.0787	0.5658	0.4291
Multimodal	3D Late Fusion	0.8439 $\pm$ 0.0735	0.8423 $\pm$ 0.0725	0.7407	0.7030
Multimodal	3D DAFT	0.8062 $\pm$ 0.0641	0.8328 $\pm$ 0.0716	0.7568	0.6988

To visually contextualize these results, the aggregated mean ROC curves across the 10 folds for the evaluated models utilizing the Top 200 features are plotted in Figure 4.3.



(a) Unimodal Baselines vs. Late Fusion (Top 200 Features)



(b) Intermediate Fusion DAFT (Top 200 Features)

Figure 4.3: Mean ROC curves across the 10-fold CV during the ablation study. (a) Compares the baseline models to the Decision-Level Late Fusion ensemble. (b) Displays the curve for the feature-level intermediate fusion DAFT architecture. Shaded regions represent ± 1 standard deviation.

The empirical findings from this ablation study indicate that restricting the tabular input to the Top 200 features was associated with improved performance for the intermediate fusion

architecture. As detailed in Table 4.5, the DAFT Test AUC increased from 0.8062 (using the full feature set) to 0.8328 (using the reduced subset). By contrast, the performance of the pure tabular XGBoost model and the Decision-Level Late Fusion ensemble remained relatively static across both feature sets. While the DAFT architecture exhibited the largest performance increase from the ablation and achieved the highest overall test Accuracy (0.7568), the unimodal tabular XGBoost model achieved the highest overall Mean Test AUC (0.8440), with an almost identical AUC achieved by the late fusion model (0.8439).

4.4 Statistical Significance and Pairwise Model Comparison

While aggregate classification metrics provide a general overview of model performance, it is necessary to determine whether the observed performance gaps are statistically significant. A pairwise statistical analysis was conducted on the CV test predictions. The evaluation consists of 3,504 MC-level predictions; however, because these MCs originate from 94 unique patients across the dataset, they are not statistically independent. To ensure valid inference, the statistical evaluation methods were specifically designed to account for this intra-patient correlation (clustering).

4.4.1 Bootstrapped ROC AUC Comparisons

To evaluate the statistical significance of the differences in Area Under the Curve (Δ AUC) between the architectures, a non-parametric, clustered, paired bootstrapping simulation was employed.

During this procedure, resampling was performed at the patient level rather than the MC level. For each of the 10,000 iterations, a complete bootstrapped cohort of 94 patients was generated by randomly sampling the original 94 unique patients with replacement. Once a patient was selected for a bootstrap iteration, all of their associated MC predictions from both models being compared were retrieved exactly as paired instances. This ensured that the paired nature of the predictions and the intra-patient clustering were strictly preserved. The 95% CIs for Δ AUC were constructed from this bootstrapped distribution. The p -value was calculated based on the proportion of bootstrap iterations where the difference in AUC crossed zero, multiplied by two for a two-tailed test. The results are summarized in Table 4.6.

Table 4.6: Clustered, paired bootstrapped pairwise comparisons of AUC performance (10,000 iterations). A p -value < 0.05 indicates a statistically significant difference. Accounting for patient-level clustering revealed no statistically significant differences between the architectures.

Model Comparison (A vs. B)	Mean Δ AUC	95% CI	p -value
XGBoost vs. DAFT	+0.0070	[−0.0262, 0.0399]	0.3306
XGBoost vs. Late Fusion	−0.0069	[−0.0166, 0.0015]	0.9414
DAFT vs. Late Fusion	−0.0139	[−0.0473, 0.0172]	0.8101

The clustered bootstrapping analysis indicates that none of the observed differences in AUC are statistically significant. It should be clarified that while the statistical bootstrapping resampled data at the patient level to account for correlation, the AUC metric itself was still calculated

using the individual MC-level predictions concatenated within each bootstrap iteration. Specifically, when comparing the pure tabular XGBoost model to the multimodal Late Fusion and DAFT pipelines, the 95% CIs included zero ($p = 0.9414$ and $p = 0.3306$, respectively). This demonstrates that, once intra-patient correlation is properly accounted for, combining the 3D visual features with explicitly defined radiomic features does not yield a statistically significant improvement in MC-level risk-ranking capability over using the radiomic features in isolation. Because the confidence intervals include zero, the statistical conclusion is that no significant differences were found between the architectures.

4.4.2 McNemar’s Test for Classification Concordance

While the ROC AUC evaluates continuous probability distributions, we also wanted to test the binary classifications of the models (e.g., $t = 0.5$). To evaluate whether the absolute error rates of the models differed significantly, McNemar’s Test for paired nominal data was utilized.

To properly account for patient-level clustering in this binary evaluation, the 3,504 individual MC predictions were aggregated. The probabilities for each patient were averaged to derive a single aggregate diagnostic risk score per patient, resulting in $N = 94$ independent paired predictions. McNemar’s Test was then applied to these patient-level classifications.

Table 4.7: McNemar’s Test results evaluating the statistical significance of disagreements in patient-level binary classifications ($N = 94, t = 0.5$). The test indicated no statistically significant differences in error rates.

Model Comparison (A vs. B)	Both Correct	Disagreements (A right / B right)	Both Wrong	p -value
XGBoost vs. DAFT	69	3 / 2	20	1.0000
XGBoost vs. Late Fusion	71	1 / 2	20	1.0000
DAFT vs. Late Fusion	70	1 / 3	20	0.6171

As detailed in Table 4.7, applying McNemar’s Test at the patient level yielded $p > 0.05$ across all model comparisons. This indicates that there is no statistically significant difference in the overall binary error rates between the evaluated architectures. Furthermore, the volume of classification disagreements between the models is extremely low at the patient level (e.g., XGBoost and Late Fusion only disagreed on 3 out of 94 patients). This shows that for our setup and study, the multimodal fusion strategies do not significantly alter the absolute number of correct binary patient diagnoses compared to the pure radiomic baseline at the default threshold.

4.5 Ablation Study: Constraining Late Fusion Meta-Learner Weights

In the primary 3D Late Fusion experiments, the Logistic Regression meta-learner assigned positive coefficients to the XGBoost predictions (e.g., +10.27 in Fold 1) and negative coefficients to the 3D ResNet-18 predictions (e.g., -1.74). While negative coefficients are a standard and mathematically valid outcome in unconstrained logistic regression, an ablation study was conducted to observe the ensemble’s predictive dynamics when forced to act strictly as a convex combination (a weighted average) of the unimodal probabilities.

To achieve this, the standard unconstrained Logistic Regression meta-classifier—which can assign negative weights—was replaced with a custom neural layer during training. The learnable

modality weights were passed through a Softmax function during the forward pass. This constraint ensured that the weights assigned to the XGBoost and 3D CNN probabilities remained strictly positive and summed to exactly 1.0.

Empirical Observations on Bounded Weights

The unimodal baselines and the constrained meta-learner were evaluated using the second private database only, and under the identical 10-fold CV framework. To evaluate the impact of this constraint, Table 4.8 compares the fold-by-fold Test AUC of this constrained model directly against the Unconstrained Late Fusion baseline.

Table 4.8: Fold-by-fold analysis of the constrained late fusion ablation study. The meta-learner weights were constrained via a Softmax function during training to be strictly positive and sum to 1.0. Across all folds, the optimizer assigned the vast majority of the weight to the tabular baseline. The Δ column compares the constrained AUC to the original unconstrained Late Fusion AUC.

Fold	Unimodal Test AUC		Learned Softmax Weights		Constrained Ensemble	
	XGBoost	3D CNN	XGBoost Weight	3D CNN Weight	Fusion Test AUC	Δ vs. Unconstrained
1	0.9386	0.6652	0.9924	0.0076	0.9384	+0.0017
2	0.8369	0.5990	0.9938	0.0062	0.8374	+0.0072
3	0.9125	0.6695	0.9939	0.0061	0.9126	+0.0014
4	0.7559	0.5906	0.9939	0.0061	0.7549	-0.0094
5	0.8158	0.6630	0.9932	0.0068	0.8155	-0.0026
6	0.7862	0.7495	0.9922	0.0078	0.7871	+0.0127
7	0.9195	0.7158	0.9939	0.0061	0.9194	-0.0021
8	0.8835	0.5887	0.9935	0.0065	0.8830	-0.0005
9	0.8858	0.6721	0.9937	0.0063	0.8857	+0.0094
10	0.7051	0.5549	0.9938	0.0062	0.7048	-0.0022
Mean	0.8440	0.6468	0.9934	0.0066	0.8442	+0.0003

As illustrated in Table 4.8, when restricted to a convex combination, the meta-learner consistently learned to assign over 99% of the weighting to the tabular XGBoost predictions and less than 1% to the 3D CNN predictions across all 10 folds.

When comparing the overall performance, the Constrained Late Fusion architecture achieved a Mean Test AUC of 0.8442 ± 0.0735 . This result is highly comparable to the Unconstrained Late Fusion baseline, which achieved a Mean Test AUC of 0.8439 ± 0.0735 .

This suggests that when the ensemble is restricted from assigning negative coefficients to the visual modality, the learned weights strongly favor the tabular model. By doing so, the constrained ensemble effectively mirrors the predictions of the pure tabular baseline, resulting in a predictive performance that closely tracks the unconstrained multimodal evaluation.

4.6 Ablation Study: Learnable Affine Modulation Weights in DAFT

In the standard DAFT architecture, the scale (α) and shift (β) vectors generated by the tabular bottleneck are applied directly to the intermediate spatial feature maps. To evaluate whether the network could benefit from dynamically gating or prioritizing one of these operations over

the other, an extended DAFT formulation was evaluated. Global, learnable scalar weights (w_α and w_β) were introduced to control the overall magnitude of the affine modulation:

$$F'_{i,c} = (1.0 + \sigma(w_\alpha)\alpha_{i,c})F_{i,c} + (\sigma(w_\beta)\beta_{i,c}) \quad (4.1)$$

The raw scalars were initialized at 0.0. By passing them through a Sigmoid activation function (σ), their initial effective weighting evaluated to exactly 0.5 (a 50% scaling factor). During training, the optimizer was permitted to independently adjust these weights to determine the relative importance of the scaling and shifting components. The experiment was conducted utilizing the best-performing configuration from the previous ablation (Top 200 radiomic features, learning rate of 1×10^{-3} , and a 50% dropout rate) under the identical 10-fold CV framework.

Analysis of Weight Dynamics and Performance

The inclusion of the learnable global scalar weights resulted in a Mean Test AUC of 0.8353. The comprehensive classification metrics for this configuration are detailed in Table 4.9.

Table 4.9: 10-fold CV test results for the DAFT architecture utilizing learnable global scalar weights for the shift and scale operations.

Mean Test AUC	Test Accuracy	Test F1-Score	Test Sensitivity	Test Specificity
0.8353 ± 0.0721	0.7208	0.6781	0.7431	0.7332

This performance is highly comparable to the standard DAFT configuration evaluated under the same conditions (Mean Test AUC of 0.8328). The addition of these global trainable parameters did not substantially alter the overall performance of the intermediate fusion pipeline.

To understand how the network utilized these parameters, the epoch-by-epoch trajectories of the Sigmoid-activated weights ($\sigma(w_\alpha)$ and $\sigma(w_\beta)$) were logged and analyzed. It is important to note that because the architecture was trained independently across a 10-fold CV framework, each fold learned its own distinct pair of scalar weights. Furthermore, while these scalar parameters are updated during training, they become fixed global constants during inference, unlike the underlying α and β vectors which are dynamically generated per patient.

Despite being trained independently, the networks exhibited a consistent behavioral pattern across all folds. During the initial phase of training (approximately epochs 1 through 20), the optimizer consistently increased the magnitude of both operations, with the shifting component peaking at approximately 0.57 and the scaling component peaking at approximately 0.52.

However, as training progressed beyond epoch 30 and the networks approached their lowest validation losses, both weights steadily decayed back toward their initialized baseline. By the time the early stopping criteria were met (typically between epochs 80 and 89), the final $\sigma(w_\alpha)$ and $\sigma(w_\beta)$ values had stabilized back to approximately 0.50 consistently across all 10 folds.

Ultimately, this consistent convergence indicates that the additional global gates stayed very close to their exact initialization values at the end of training. What could be concluded is that introducing these global weighting parameters did not provide a substantial advantage or significantly change the model’s performance on this dataset.

Chapter 5

Discussion

The findings across both experimental pipelines provide insights into the behavior of multimodal fusion architectures and the representational capacity of tabular features—specifically radiomic features—in medical image analysis.

5.1 Analysis of Performance Across Pipelines

In the first pipeline utilizing 2D clinical mammography (EMBED dataset), the visual baselines achieved limited performance. The most likely explanation for this is the very limited number of images, which is not sufficient for the vision networks to learn properly. Another possible explanation for this difficulty is the inherent tissue superimposition in 2D imaging, which may complicate the extraction of reliable diagnostic geometries from small ROIs. In this setting, the intermediate fusion approach (DAFT) yielded higher AUCs than the unimodal baselines, suggesting that integrating clinical tabular variables into the intermediate layers might provide useful contextual information when visual features are constrained.

In contrast, the second pipeline evaluated high-resolution 3D Micro-CT imaging of ex-vivo biopsy specimens paired with extracted radiomic descriptors [18]. Here, the tabular model (XGBoost) achieved a higher baseline performance compared to the 3D CNNs. It is important to note that the 2D and 3D pipelines represent fundamentally different tasks, datasets, and tabular feature spaces (standard clinical demographics versus hundreds of engineered mathematical descriptors). Therefore, the performance differences cannot be solely attributed to the dimensionality of the images. However, within the 3D pipeline itself, the results suggest that for sparse and small structures like individual MCs, mathematically defined radiomic descriptors may be more effective at capturing diagnostic information than the spatial features extracted implicitly by the evaluated 3D CNNs.

5.2 Mechanisms of Multimodal Fusion and Meta-Learner Dynamics

In the 3D pipeline, the Late Fusion ensemble achieved an overall Mean Test AUC comparable to the pure tabular XGBoost model. Analysis of the unconstrained Logistic Regression meta-learner revealed that it assigned a large positive coefficient to the XGBoost probabilities and a negative coefficient to the 3D CNN probabilities.

While this might initially appear counterintuitive, studies show that negative coefficients are a standard outcome in logistic regression when features are correlated or when calibration adjustments are implicitly made by the model [30, 31]. It reflects how the CNN probability contributes conditionally within the fitted model. This dynamic was further contextualized in the ablation study: when the meta-learner was constrained to act as a strict convex combination (forcing weights to be positive and sum to 1.0), the optimizer assigned over 99% of the weighting to XGBoost. This indicates that, under these constraints, the deep visual features act primarily as minor modulators rather than equivalent drivers of the final prediction.

5.3 Statistical Significance and Information Redundancy

The pairwise statistical analysis evaluated whether the observed performance metrics represented statistically significant differences. Utilizing a patient-level clustered bootstrapping simulation to account for intra-patient correlation, the comparisons revealed no statistically significant differences in the continuous risk-ranking metric (ΔAUC) between the optimized multimodal architectures and the pure tabular XGBoost baseline. Specifically, the CIs for the comparisons between XGBoost, DAFT, and Late Fusion all included zero.

This indicates that integrating 3D visual features with radiomic features did not yield a statistically significant improvement over utilizing the radiomic features in isolation. The bootstrap analysis solely demonstrates that no significant differences were detected, rather than proving the models are definitively equivalent. One possible explanation for this outcome is a high degree of information redundancy, where the visual representations extracted by the 3D CNNs do not provide complementary diagnostic information beyond what is already captured by the curated radiomic feature set. It is also possible that complementary visual information does exist within the Micro-CT scans, but the specific dataset size, chosen network architectures, or current fusion strategies were simply unable to effectively extract and exploit it. However, these remain speculations that are not proven by these tests.

While the AUC evaluations during the main pipeline were performed at the individual MC level (with resampling clustered by patient), it is important to again clearly separate this from the binary classification evaluation, which was strictly conducted at the aggregated patient level. After averaging the individual MC probabilities to derive a single diagnostic risk score per patient, McNemar’s Test did not reveal any statistically significant differences in absolute error rates across any of the architecture pairings at the standard $t = 0.5$ threshold. The volume of classification disagreements between the models was exceptionally low at this patient level. This indicates that, using patient-level predictions, the multimodal fusion strategies did not yield a statistically significant difference in absolute binary diagnostic accuracy at the patient level compared to the pure radiomic baseline in this ex-vivo experimental setting.

5.4 Comparison with Related Work

The findings from this thesis can be contextualized within the broader landscape of multimodal breast cancer-related research using DAFT and radiomic features. Regarding the application of the DAFT architecture, previous research by Nguyen et al. [33] experimentally evaluated DAFT for breast cancer risk prediction using 2D sequential mammograms and radiomic features. In their study, they found that while DAFT was a viable method, they ultimately transitioned to a simpler, lightweight approach that concatenated all features into a fully-connected layer, as this less complex architecture yielded their best overall predictive performance [33]. Other than the mentioned study, and at the time of writing, no other study was found using DAFT

with radiomic features, and no other study was found using DAFT with radiomics as their main fusion structure. We note that there is no direct performance comparison between our results and Nguyen et al., as everything from the dataset to the setup and architectures differs, and we only point to similarities based on both studies using DAFT and radiomic features.

Furthermore, within the specific context of the 3D Micro-CT dataset evaluated in this thesis, previous studies by Brahimetaj et al. [18, 25] have demonstrated that radiomic features can effectively capture the morphological complexity of individual MCs and improve the prediction of malignancy. Our multimodal evaluation aligns with this finding, with the pure radiomic XG-Boost model maintaining the highest overall AUC. However, direct numerical comparisons of performance metrics (e.g., absolute AUCs) between this thesis and prior publications on this dataset should be interpreted with caution. The experimental designs are not strictly identical; differences in cross-validation splitting strategies (such as our implementation of patient-level clustered 10-fold CV to strictly prevent data leakage), the specific isolated subsets of MCs evaluated, and the chosen thresholding configurations inherently limit the validity of direct, one-to-one metric comparisons. Nonetheless, the overarching conclusion of tabular models using radiomic features being at least on par with the more complex fusion networks seems to be consistent across these papers and our work.

Finally, regarding the specific application of the DAFT architecture to 3D imaging of individual breast microcalcifications, a review of the current literature yields no prior studies. To the best of our knowledge, at the time of writing, there are no published papers or preprints evaluating DAFT on individual 3D microcalcifications. Consequently, this thesis serves as an initial exploration into this specific application.

5.5 Experimental Limitations

A methodological limitation in this study stems from the training dynamics inherent to k-fold CV when paired with early stopping. To mitigate overfitting, an early stopping mechanism was implemented based on the validation loss. While this is a standard regularization technique, it introduces variance in model convergence across the folds, as differing patient distributions lead to different optimal stopping epochs.

Consequently, the aggregated Mean Test AUC reported in this study represents an ensemble average of networks trained for varying durations, rather than a single model trained for a definitive number of epochs. For the development of a final, unified model, this variance can be addressed by deriving a fixed training duration from the CV runs (such as the median stopping epoch) or by retaining an independent validation set to explicitly govern the early stopping of the final network.

A fundamental limitation of this study is the constrained sample size of both datasets, particularly the 2D EMBED cohort. In deep learning and multimodal research, significantly larger datasets are generally required to ensure robust feature extraction and reliable model generalization. Furthermore, the absence of an independent, external validation cohort restricts the broader applicability of these findings. Consequently, the conclusion that explicitly defined radiomic features establish the optimal predictive baseline must currently be interpreted within the context of this specific ex-vivo dataset and experimental configuration, rather than as a generalized claim.

Chapter 6

Conclusion and Future Work

This thesis evaluated multimodal data fusion strategies for the classification of breast MCs. The study investigated decision-level (late) fusion and feature-level intermediate fusion via DAFT across two experimental pipelines using two different datasets. To the best of our knowledge, this work establishes the first baseline for DAFT’s implementation within a high-resolution 3D environment utilizing mathematically engineered radiomic features. The models were evaluated using patient-level CV to account for intra-patient correlation and to provide a consistent comparison against unimodal baselines.

In the first pipeline, which evaluated 2D clinical mammography, the inclusion of basic clinical tabular data via the DAFT architecture yielded performance improvements over the unimodal visual baselines. These results indicate that intermediate feature-level conditioning can provide useful context to DL models when visual feature extraction is challenging. The open-source part of the dataset, however, was very limited. We attempted to expand to using all available images regardless of whether they contained MCs or not, but that did not provide a substantial increase to the dataset size and consequently did not change the results obtained.

The second pipeline explored the integration of high-resolution 3D Micro-CT images with mathematically engineered radiomic features. In this ex-vivo setting, the unimodal XGBoost model trained exclusively on radiomic features established a strong predictive baseline. The clustered statistical analysis revealed no statistically significant differences in performance between the pure radiomic model, the intermediate DAFT architecture, and the Late Fusion ensemble. While the DAFT architecture successfully integrated the modalities and improved upon the image-only networks, it did not significantly exceed the performance of the radiomic features alone. Furthermore, the ablation study reducing the radiomic feature space to a smaller subset of variables showed a tendency towards improvement of the performance of the DAFT model, but no statistical analysis was done to prove a statistical improvement to performance by using the mentioned reduced subset of features.

The findings from this ex-vivo study indicate that for the classification of 3D MC morphology, deep convolutional architectures do not provide a statistically significant improvement over explicitly defined radiomic descriptors. Because the multimodal fusion models performed comparably to the pure radiomic baseline without significantly altering the volume of correct binary classifications, a model utilizing solely radiomic features represents an efficient and effective approach for this specific task.

As a continuation of this research, several avenues for future work can be explored to build upon the established study. First, to properly assess the generalization capabilities of multimodal

architectures on clinical 2D/3D imaging, the same experimental pipeline should be evaluated on a significantly larger and fully annotated dataset pairing tabular data with imaging, which could overcome the sample size limitations and the overfitting challenges encountered in this work. Second, future studies could investigate alternative state-of-the-art multimodal fusion networks to determine if different architectural mechanisms can better exploit potential complementary information between the visual and tabular domains. Another avenue to explore is conducting a comparative analysis to evaluate model performance on full volumetric patches containing complete clusters of MCs versus the isolated, individual MCs using a similar experimental pipeline. Finally, a next step that could address one of the primary limitations of this study is to evaluate these architectures on a dataset of medical images and tabular data in the field of breast microcalcifications that are, in fact, complementary, specifically one featuring a strong image-only baseline, as the image-only baseline was consistently lagging behind the tabular data-only models in our experiments. Using fusion networks where both modalities are being exploited to improve the classification results of the model could be the best way forward from this research.

Appendix A

Complete Feature Lists

This appendix provides the complete list of features for both the EMBED and private 3D Micro-CT datasets.

A.1 EMBED Dataset Features

Table A.1 and Table A.2 detail the complete list of all the clinical, metadata, and DICOM features from the EMBED dataset used in the first pipeline.

Table A.1: Complete list of clinical and metadata tabular features from the EMBED dataset.

Feature	Feature
massshape	massmargin
massdens	calcfind
calcdistri	calcnnumber
otherfind	implanfind
consistent	side
size	location
depth	distance
numfind	asses
recc	stable
new	changed
loc_num	tech_init
init	proccode
desc	vtype
tissueden	case
type	technique
biopsite	biop_loc
bcomp	path_loc
diag_out	surgery
lymphsurg	surg_loc
pocomp	ltcomp
bside	path1
path2	path3
path4	path5
path6	path7
path8	path9
path10	concord
hgrade	tnmpt
tnmpn	tnmm
tnmdesc	tnmr
stage	loc
bdepth	bdistance
focality	nfocal
specszie	specszie2

Continued on next page

Table A.1 – continued from previous page

Feature	Feature
specsize3	dcissize
invsiz	superior
inferior	anterior
posterior	medial
lateral	specinteg
specnum	specembed
est	estp
her2	fish
ki67	extracap
methodevl	snode_rem
node_rem	node_pos
macrometa	micrometa
isocell	largedp
eic	GENDER_DESC
ETHNICITY_DESC	ETHNIC_GROUP_DESC
MARITAL_STATUS_DESC	ENCOUNTER_QTY
age_at_study	empi_anon
acc_anon	study_date_anon
sdate_anon	procddate_anon
pdate_anon	cohort_num
path_group	path_severity
total_L_find	total_R_find
first_3_zip	

Table A.2: Complete list of metadata features from the EMBED dataset.

Feature	Feature
AcquisitionContextSequence	AcquisitionTime
0_AnatomicRegionSequence_CodeMeaning	0_AnatomicRegionSequence_CodeValue
0_AnatomicRegionSequence_CodingSchemeDesignator	AnodeTargetMaterial
BitsAllocated	BitsStored
BodyPartExamined	BodyPartThickness
BreastImplantPresent	BurnedInAnnotation
Columns	CompressionForce
ContentTime	DetectorBinning
DetectorConditionsNominalFlag	DetectorTemperature
DetectorType	DistanceSourceToDetector
DistanceSourceToPatient	EntranceDose
EntranceDoseInmGy	EstimatedRadiographicMagnificationFactor
Exposure	ExposureControlMode
ExposureControlModeDescription	ExposureInuAs
ExposureTime	ExposureTimeInuS
FieldOfViewHorizontalFlip	FieldOfViewOrigin
FieldOfViewRotation	FilterMaterial
FilterThicknessMaximum	FilterThicknessMinimum
FilterType	FocalSpots
Grid	HalfValueLayer
HighBit	ImageLaterality
ImageType	ImagerPixelSpacing
ImagesInAcquisition	InstanceNumber
KVP	LossyImageCompression
Manufacturer	ManufacturerModelName
Modality	OrganDose
OrganExposed	PaddleDescription
PatientOrientation	PatientSex
PerformedProcedureStepDescription	PerformedProcedureStepStartTime
0_PerformedProtocolCodeSequence_CodeMeaning	0_PerformedProtocolCodeSequence_CodeValue
0_PerformedProtocolCodeSequence_	PhotometricInterpretation
CodingSchemeDesignator	
PixelIntensityRelationship	PixelIntensityRelationshipSign
PixelPaddingValue	PixelRepresentation
PixelSpacing	PositionerPrimaryAngle
PositionerType	PresentationIntentType
PresentationLUTShape	0_ProcedureCodeSequence_CodeMeaning
0_ProcedureCodeSequence_CodeValue	0_ProcedureCodeSequence_CodingSchemeDesignator
0_ProcedureCodeSequence_CodingSchemeVersion	ProtocolName

Continued on next page

Table A.2 – continued from previous page

Feature	Feature
QualityContrastImage	RelativeXRayExposure
RescaleIntercept	RescaleSlope
RescaleType	Rows
SamplesPerPixel	SeriesDescription
SeriesNumber	SeriesTime
0_SourceImageSequence_0_	0_SourceImageSequence_0_
PurposeOfReferenceCodeSequence_CodeMeaning	PurposeOfReferenceCodeSequence_CodeValue
0_SourceImageSequence_0_	0_SourceImageSequence_SpatialLocationsPreserved
PurposeOfReferenceCodeSequence_	
CodingSchemeDesignator	
SpecificCharacterSet	StudyDescription
StudyTime	TimeOfLastDetectorCalibration
0_ViewCodeSequence_CodeMeaning	0_ViewCodeSequence_CodeValue
0_ViewCodeSequence_CodingSchemeDesignator	0_ViewCodeSequence_ViewModifierCodeSequence
ViewPosition	WindowCenter
WindowWidth	XRayTubeCurrent
PositionerPrimaryAngleDirection	PixelPaddingRangeLimit
0_ViewCodeSequence_0_ViewModifierCodeSequence_	0_ViewCodeSequence_0_ViewModifierCodeSequence_
CodeMeaning	CodeValue
0_ViewCodeSequence_0_ViewModifierCodeSequence_	AcquisitionDeviceProcessingCode
CodingSchemeDesignator	
AcquisitionDeviceProcessingDescription	DerivationDescription
DetectorConfiguration	FieldOfViewDimensions
FieldOfViewShape	Laterality
Sensitivity	ConversionType
0_RelatedSeriesSequence_0_	0_RelatedSeriesSequence_0_
PurposeOfReferenceCodeSequence_CodeMeaning	PurposeOfReferenceCodeSequence_CodeValue
0_RelatedSeriesSequence_0_	CollimatorLeftVerticalEdge
PurposeOfReferenceCodeSequence_	
CodingSchemeDesignator	
CollimatorLowerHorizontalEdge	CollimatorRightVerticalEdge
CollimatorShape	CollimatorUpperHorizontalEdge
CommentsOnRadiationDose	DetectorActiveDimensions
DetectorActiveShape	DetectorElementPhysicalSize
DetectorElementSpacing	DistanceSourceToEntrance
ExposureStatus	0_IconImageSequence_BitsAllocated
0_IconImageSequence_BitsStored	0_IconImageSequence_Columns
0_IconImageSequence_HighBit	0_IconImageSequence_PhotometricInterpretation
0_IconImageSequence_PixelRepresentation	0_IconImageSequence_Rows
0_IconImageSequence_SamplesPerPixel	VOILUTFunction
WindowCenterWidthExplanation	DetectorDescription
empi_anon	acc_anon
study_date_anon	cohort_num
LateralityDeriveFlag	ImageLateralityFinal
FinalImageType	spot_mag
anon_dicom_path	ROI_coords
num_ROI	

A.2 Ablation Study: Reduced Radiomic Features (Top 200)

Table A.3 lists the reduced subset of 200 radiomic features extracted from the 3D Micro-CT dataset that were utilized in the targeted ablation study. These features were identified as the most discriminative subset based on recursive feature elimination.

Table A.3: Subset of 200 radiomic features utilized in the ablation study.

Feature	Feature
wavelet-LHL_glcM_DifferenceAverage	wavelet-LHL_gldM_LargeDependenceEmphasis
wavelet-LHL_glrM_LongRunHighGrayLevelEmphasis	wavelet-LHL_glcM_Correlation
wavelet-LHL_glcM_Contrast	wavelet-LHL_glrM_RunEntropy
wavelet-LHL_glcM_MaximumProbability	wavelet-LHL_glrM_ShortRunEmphasis

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Table A.3 – continued from previous page

Feature	Feature
wavelet-LHL_glrml_LongRunEmphasis	wavelet-LHL_glrml_LongRunLowGrayLevelEmphasis
wavelet-LHL_gldm_	wavelet-LHL_gldm_SmallDependenceEmphasis
SmallDependenceLowGrayLevelEmphasis	
wavelet-LHL_glszm_LowGrayLevelZoneEmphasis	wavelet-LHL_gldm_DependenceVariance
wavelet-LHL_glszm_GrayLevelVariance	wavelet-LHL_glszm_GrayLevelNonUniformity
wavelet-LHL_glszm_SmallAreaEmphasis	wavelet-LHL_gldm_
	DependenceNonUniformityNormalized
wavelet-LHL_ngtdm_Complexity	wavelet-LHL_glszm_
	SizeZoneNonUniformityNormalized
wavelet-LHL_glszm_	wavelet-LHL_glszm_ZonePercentage
SmallAreaLowGrayLevelEmphasis	
wavelet-LHL_glszm_	wavelet-LHL_ngtdm_Strength
GrayLevelNonUniformityNormalized	
wavelet-LHL_glszm_	original_gldm_ClusterShade
SmallAreaHighGrayLevelEmphasis	
wavelet-HHL_gldm_Imc1	original_gldm_ClusterProminence
logarithm_gldm_ClusterShade	original_gldm_Correlation
wavelet2-HLL_gldm_Imc1	original_glrml_RunEntropy
logarithm_gldm_ClusterProminence	squareroot_gldm_ClusterShade
original_gldm_Autocorrelation	wavelet-HLH_gldm_Imc1
wavelet2-LHH_firstorder_Skewness	wavelet2-LHL_gldm_Imc1
wavelet2-HLH_firstorder_Skewness	wavelet2-LHL_gldm_Correlation
wavelet2-HHL_firstorder_Skewness	wavelet-HLL_gldm_Imc1
original_gldm_	log-sigma-2-5-mm-3D_gldm_Correlation
LargeDependenceLowGrayLevelEmphasis	
log-sigma-2-5-mm-3D_gldm_ClusterShade	log-sigma-2-5-mm-3D_gldm_Imc1
original_glszm_GrayLevelVariance	log-sigma-2-5-mm-3D_glszm_LargeAreaEmphasis
original_ngtdm_Complexity	log-sigma-2-5-mm-3D_glrml_
	LongRunLowGrayLevelEmphasis
wavelet-LH_gldm_DifferenceAverage	wavelet3-LHL_gldm_Imc1
original_glrml_LongRunLowGrayLevelEmphasis	wavelet2-LHL_gldm_DifferenceAverage
wavelet2-HLL_gldm_Correlation	wavelet2-HHL_firstorder_Range
log-sigma-2-5-mm-3D_gldm_ClusterProminence	log-sigma-2-5-mm-3D_glrml_LongRunEmphasis
wavelet-HLH_firstorder_Skewness	wavelet-LHH_gldm_ClusterShade
wavelet3-LLH_gldm_Imc1	wavelet-HLL_gldm_DifferenceAverage
wavelet-LHH_firstorder_Skewness	wavelet-HHH_firstorder_Skewness
wavelet3-LLH_gldm_Correlation	wavelet-HLL_gldm_Correlation
wavelet2-HLH_firstorder_Range	wavelet-LH_glrml_LongRunLowGrayLevelEmphasis
wavelet2-HLH_glszm_GrayLevelNonUniformity	wavelet2-HHL_firstorder_Median
wavelet-HHL_firstorder_Range	wavelet2-HLH_firstorder_Mean
wavelet-HLH_firstorder_Range	logarithm_gldm_Correlation
wavelet2-HLH_firstorder_Median	wavelet-LHH_firstorder_MeanAbsoluteDeviation
wavelet-HLH_firstorder_MeanAbsoluteDeviation	wavelet-HHL_gldm_Correlation
wavelet2-LLH_gldm_Imc1	wavelet-LH_gldm_LargeDependenceEmphasis
wavelet2-LLH_gldm_DifferenceAverage	logarithm_glrml_LongRunLowGrayLevelEmphasis
wavelet-LHH_gldm_Autocorrelation	wavelet-LLH_gldm_Correlation
original_glszm_LargeAreaLowGrayLevelEmphasis	squareroot_gldm_Correlation
wavelet-HHL_gldm_Autocorrelation	wavelet2-LHH_gldm_Autocorrelation
original_glrml_RunLengthNonUniformity	wavelet-HHL_firstorder_MeanAbsoluteDeviation
wavelet-LLH_glrml_LongRunEmphasis	wavelet2-LLL_gldm_Correlation
original_glszm_LargeAreaEmphasis	wavelet-HHL_glszm_LargeAreaEmphasis
original_glszm_SizeZoneNonUniformity	wavelet-LLH_gldm_Imc1
wavelet2-LHL_gldm_Autocorrelation	log-sigma-2-5-mm-3D_gldm_DependenceVariance
log-sigma-2-5-mm-3D_glszm_ZoneEntropy	wavelet-HHL_gldm_ClusterTendency
wavelet-HLH_glszm_LargeAreaEmphasis	wavelet3-LHL_gldm_Autocorrelation
square_glszm_SizeZoneNonUniformity	original_glszm_GrayLevelNonUniformity
wavelet-HHH_firstorder_MeanAbsoluteDeviation	logarithm_gldm_Imc1
wavelet-LHH_gldm_JointEnergy	wavelet2-HHL_firstorder_Mean
wavelet-LHH_gldm_ClusterProminence	square_gldm_ClusterProminence
log-sigma-2-5-mm-3D_gldm_Autocorrelation	wavelet-HHH_gldm_JointEnergy
squareroot_gldm_Imc2	square_glrml_RunLengthNonUniformity
square_gldm_Correlation	wavelet2-LLH_gldm_Correlation
log-sigma-2-5-mm-3D_glszm_SizeZoneNonUniformity	original_glszm_HighGrayLevelZoneEmphasis
wavelet-HLH_gldm_JointEnergy	wavelet-LHL_gldm_Autocorrelation
wavelet-HLH_gldm_Autocorrelation	log-sigma-2-5-mm-3D_glrml_RunEntropy
wavelet2-LLL_gldm_Imc2	original_gldm_Imc1
wavelet3-LHL_gldm_Correlation	log-sigma-2-5-mm-3D_glszm_GrayLevelNonUniformity
wavelet3-HHL_gldm_Autocorrelation	wavelet-HLH_gldm_ClusterTendency
log-sigma-2-5-mm-3D_ngtdm_Complexity	log-sigma-2-5-mm-3D_glszm_ZonePercentage

Continued on next page

Table A.3 – continued from previous page

Feature	Feature
original_gldm_DependenceVariance	wavelet-HLH_gldm_ClusterProminence
wavelet3-HHH_gldm_Autocorrelation	wavelet-LLH_gldm_Autocorrelation
wavelet-HHL_gldm_ClusterProminence	wavelet-LHH_gldm_ClusterTendency
original_gldm_ShortRunHighGrayLevelEmphasis	log-sigma-2-5-mm-3D_gldm_DependenceNonUniformityNormalized
wavelet-HHL_gldm_JointEnergy	logarithm_gldm_RunLengthNonUniformity
wavelet3-LHH_gldm_Autocorrelation	original_glszm_SmallAreaHighGrayLevelEmphasis
wavelet2-HLH_gldm_Autocorrelation	wavelet3-HLL_gldm_Correlation
wavelet2-HHL_gldm_Autocorrelation	wavelet3-LLL_gldm_Imc2
log-sigma-2-5-mm-3D_gldm_ShortRunEmphasis	wavelet3-HLL_gldm_Imc1
wavelet2-HLH_firstorder_Minimum	wavelet2-HLL_gldm_DifferenceAverage
wavelet3-HHL_firstorder_Skewness	wavelet3-HLH_gldm_Autocorrelation
log-sigma-2-5-mm-3D_glszm_SizeZoneNonUniformityNormalized	wavelet2-HHL_firstorder_Minimum
original_gldm_Imc2	log-sigma-2-5-mm-3D_gldm_RunPercentage
wavelet3-LLL_gldm_Correlation	wavelet2-HLH_firstorder_Maximum
wavelet-LHH_gldm_Contrast	wavelet-HLL_gldm_RunEntropy
wavelet2-HHH_firstorder_Skewness	log-sigma-2-5-mm-3D_gldm_SmallDependenceEmphasis
wavelet-HLH_gldm_Contrast	original_gldm_SmallDependenceHighGrayLevelEmphasis
wavelet3-HLH_firstorder_Skewness	wavelet-HLL_gldm_Autocorrelation
original_glszm_ZonePercentage	wavelet-HHL_gldm_Contrast
wavelet3-LLH_gldm_Autocorrelation	log-sigma-2-5-mm-3D_glszm_SmallAreaEmphasis
original_gldm_LargeDependenceHighGrayLevelEmphasis	wavelet2-HHH_gldm_Autocorrelation
logarithm_gldm_Imc2	wavelet-HLL_gldm_ClusterShade
wavelet-HHH_gldm_ClusterProminence	log-sigma-2-5-mm-3D_gldm_RunLengthNonUniformityNormalized
wavelet2-HHH_gldm_ClusterProminence	wavelet-HLH_gldm_DifferenceEntropy
wavelet-HLL_gldm_LongRunLowGrayLevelEmphasis	wavelet-HHL_gldm_Imc2
log-sigma-2-5-mm-3D_glszm_GrayLevelNonUniformityNormalized	log-sigma-2-5-mm-3D_gldm_ShortRunHighGrayLevelEmphasis
wavelet-LLH_gldm_RunEntropy	wavelet-LHH_glszm_ZoneVariance
wavelet2-LHH_gldm_ClusterProminence	original_glszm_ZoneEntropy
wavelet2-HHL_firstorder_Maximum	original_glszm_SizeZoneNonUniformityNormalized
original_glszm_GrayLevelNonUniformityNormalized	wavelet-LLH_gldm_JointEnergy
wavelet-HHL_gldm_DifferenceAverage	log-sigma-2-5-mm-3D_gldm_GrayLevelNonUniformityNormalized
wavelet-LHL_firstorder_InterquartileRange	original_gldm_GrayLevelNonUniformityNormalized
log-sigma-2-5-mm-3D_gldm_SmallDependenceHighGrayLevelEmphasis	original_gldm_DependenceNonUniformityNormalized
wavelet2-HHL_gldm_ClusterProminence	wavelet2-LLH_gldm_Autocorrelation
square_gldm_Imc2	wavelet-LHH_gldm_Idmn
wavelet3-HHH_gldm_Autocorrelation	wavelet-LLH_gldm_ClusterTendency
wavelet2-HLH_gldm_ClusterProminence	wavelet-HHH_gldm_ClusterShade
wavelet2-LHH_gldm_LowGrayLevelRunEmphasis	wavelet2-LHL_firstorder_Kurtosis
log-sigma-2-5-mm-3D_firstorder_Entropy	logarithm_gldm_LongRunLowGrayLevelEmphasis
wavelet2-HLL_gldm_MCC	wavelet3-HLL_firstorder_Maximum
wavelet-LLH_gldm_DependenceNonUniformityNormalized	wavelet-HLL_firstorder_InterquartileRange
wavelet-HLL_gldm_MaximumProbability	square_firstorder_10Percentile
wavelet-LHL_firstorder_Skewness	wavelet3-LHL_glszm_ZoneEntropy

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